

When Wholesalers Join the Cartel: Vertical Coordination in Gasoline Markets

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Abstract

Vertical coordination is often viewed as potentially efficiency-enhancing because it can align incentives across production stages. This paper shows that, in collusive environments, the same vertical links can amplify market power. I study eight Brazilian gasoline cartels using station-level weekly retail and wholesale prices, station characteristics, and legal records from CADE. A two-step matching difference-in-differences design shows that vertical cartels generate substantially larger increases in retail prices and retail markups than horizontal cartels, with weaker but consistent evidence of larger effects on wholesale markups. Small stations are especially affected under vertical cartels. I rationalize these findings with a model of capacity-constrained retail collusion with upstream participation. Horizontal retail cartels are limited by fringe capacity and the participation incentives of small retailers. Vertical cartels weaken the outside option of non-participants through discriminatory wholesale prices, induce full downstream participation, and divide surplus through wholesale terms. A calibrated exercise reproduces the ranking observed in the data. The findings imply that vertical participation can make cartels substantially more harmful to consumers.

Keywords: Collusion, Vertical Contracting, Fuel Retail, Antitrust enforcement, Nash Bargaining

JEL codes: L41, L42, K21, L71, C23

1 Introduction

Vertical integration and vertical coordination have long occupied an ambiguous place in antitrust analysis. Classic contributions emphasize that closer alignment

between upstream and downstream firms may improve pricing efficiency and, in some environments, reduce final prices (Spengler, 1950; Rey and Tirole, 2007). By eliminating successive markups or facilitating coordination within the firm, vertical integration is often viewed as benign or even welfare-enhancing relative to horizontal concentration.

At the same time, a large body of work recognizes that vertical arrangements can generate competitive harm. Vertical relationships may be used to foreclose rivals, soften downstream competition, or facilitate exclusion. More recently, theoretical and empirical research has shown that vertical links can play a central role in sustaining collusion, particularly in hub-and-spoke settings in which upstream firms help coordinate or discipline downstream rivals (Jullien and Rey, 2007; Van Cayseele and Miegielsen, 2013; Harrington, 2017; Clark et al., 2024; Chaves and Duarte, 2025).

What remains less understood is how vertical participation changes the *magnitude* and *organization* of collusive outcomes. The literature has largely focused on whether vertical relationships make collusion easier to sustain. Less is known about whether upstream participation makes cartels more harmful once collusion occurs. Do vertical cartels generate larger retail-price effects than purely horizontal retail cartels? Do they affect wholesale margins? And do they change which downstream firms participate in the cartel?

This paper answers these questions using evidence from Brazilian gasoline retail markets. I compare horizontal cartels, in which collusion occurs only among retailers, with vertical cartels, in which wholesalers also participate in the collusive agreement. The empirical results show that vertical cartels generate substantially larger effects on retail prices and retail markups than horizontal cartels. The evidence on wholesale markups is less precise, but it is consistent with wholesale margins responding more strongly in vertical-cartel cases. Firm-level heterogeneity further shows that small station groups are more affected under vertical cartels, suggesting that upstream participation changes the participation constraints that limit purely horizontal downstream collusion.

The paper proceeds in four steps. First, using CADE legal records from eight cartel cases (four vertical and four horizontal), I provide evidence on the mechanisms and internal organization of horizontal and vertical cartels. The goal of this step is to document how cartel organization differs across cases and to show how wholesalers can cooperate with retailers in vertical cartels. I identify two features that distinguish vertical cartels from purely horizontal retail cartels. The first is the active participation of wholesalers in pricing decisions: wholesalers attend meetings with retailers and maintain direct communication about the prices to be charged by

the cartel. The second is the use of vertical relationships to discipline retailers that deviate from the agreement. In particular, wholesalers can punish deviating stations by charging higher wholesale prices or offering less favorable contractual terms.

Second, I assemble a station-level panel for the Brazilian gasoline market that combines weekly retail and wholesale prices, station characteristics, and detailed legal records from CADE, the Brazilian antitrust authority. Using a two-step matching difference-in-differences design, I first compare cartel-affected markets to matched control markets and then compare treated stations to observationally similar stations in matched control cities. The city-level estimates show that vertical cartels generate larger increases in retail prices and retail markups than horizontal cartels. The firm-level analysis then sheds light on how these effects are distributed across stations. In vertical-cartel cases, small stations experience retail-price increases similar to those of larger stations, suggesting a relatively homogeneous pricing response. However, small stations experience smaller increases in retail markups because they also face higher wholesale prices. This pattern is consistent with the legal evidence that wholesalers use wholesale terms to discipline or pressure retailers within the cartel.

Third, I develop a theoretical model to connect the mechanisms documented in the CADE legal evidence with the patterns found in the reduced-form estimates. The model builds on the capacity-constrained retail-cartel framework of Bos and Harrington (2010) and adds an upstream wholesale tier. In a horizontal retail cartel, some retailers may remain outside the agreement and form a competitive fringe. This fringe limits residual demand and constrains the maximum sustainable cartel price. When wholesalers participate, however, they can discriminate between cartel insiders and outsiders through wholesale prices. By raising the input price faced by non-participants, vertical coordination weakens the outside option of fringe retailers and induces broader downstream participation. The model therefore links the evidence on wholesale-price discipline to the empirical finding that vertical cartels generate larger retail-price effects and that small stations face higher wholesale prices.

The model also clarifies how vertical coordination differs from separate upstream and downstream collusion. If wholesalers and retailers form independent horizontal cartels, the market features two separate collusive margins. This double-horizontal structure may generate high prices, but it is inefficient from the perspective of joint industry profits because the two tiers do not coordinate over the final retail price. Vertical coordination avoids this inefficiency by choosing the retail price jointly and using the insider wholesale price to divide surplus between wholesalers and retailers. Thus, the same vertical link that can eliminate double marginalization in competitive environments can amplify consumer harm in collusive environments.

Fourth, I use the model to conduct a calibrated numerical exercise. The exercise is not a structural estimation and is not intended to match the magnitude of the reduced-form estimates. Instead, it asks whether the economic forces emphasized by the model can reproduce the qualitative ranking observed in the data. Under a transparent parameterization disciplined by observable market characteristics, the calibrated model predicts that vertical cartels generate larger retail-price, retail-markup, and wholesale-markup effects than retail-horizontal cartels. The calibration also shows that market primitives affect cartel organization: vertical coordination is more likely when demand is relatively inelastic, while retail-horizontal collusion becomes more attractive when demand is sufficiently elastic that monopoly pricing implies large quantity losses.

The paper contributes to three literatures. First, it contributes to the empirical literature on cartels by showing that the vertical organization of collusion matters for the magnitude of overcharges and the distribution of effects across firms (Asker, 2010; Genesove and Mullin, 2001). More specifically, the paper is related to Holler and Rickert (2022), who study how resale price maintenance affects the stability and effectiveness of an upstream cartel. Their question is whether vertical restraints help producers sustain higher consumer prices by aligning retailer incentives. My paper asks a complementary question: whether wholesaler participation makes cartels more harmful than purely horizontal retail cartels. The mechanisms differ. In Holler and Rickert (2022), resale price maintenance prevents retailers from undermining upstream collusion, although loss-leading incentives can still destabilize the cartel. In my setting, wholesalers actively participate by using wholesale prices to discipline deviating or non-participating retailers. Accordingly, their main finding is that resale price maintenance increases cartel effectiveness and duration, while my results show that vertical participation amplifies retail-price and markup effects relative to horizontal collusion.

Second, it contributes to the literature on vertical restraints and hub-and-spoke collusion by providing evidence that upstream participation can do more than facilitate coordination: it can change cartel composition and increase the harm generated by collusion (Jullien and Rey, 2007; Van Cayseele and Miegielsen, 2013; Harrington, 2017; Clark et al., 2024; Chaves and Duarte, 2025). More specifically, the paper is related to Clark et al. (2024), who also study collusion involving firms at different levels of the supply chain. My paper differs by focusing on a mechanism that is better suited to gasoline retail markets. While Clark et al. (2024) emphasize supplier-retailer contracting frictions and asymmetries created by main-supplier relationships, I emphasize capacity constraints, fringe participation, and discriminatory wholesale prices. In my setting, wholesalers amplify collusion by weakening the outside option

of retailers that would otherwise remain outside a horizontal cartel.

The paper is also closely related to Chaves and Duarte (2025), since both papers study collusion in Brazilian gasoline markets. The two papers differ in scope and mechanism. While Chaves and Duarte (2025) analyze a single cartel case, I study eight cartel cases and use variation across horizontal and vertical cartels to estimate how upstream participation changes cartel outcomes. The mechanism also differs. Chaves and Duarte (2025) emphasize the role of exclusive supply agreements, through which wholesalers incompletely transmit cost changes to affiliated retailers. In contrast, my paper focuses on wholesalers' active participation in the cartel: wholesalers help sustain downstream collusion by using wholesale prices to discipline retailers that would otherwise deviate or remain outside the agreement.

Third, the paper connects the study of cartel organization to the literature on vertical bargaining by showing how wholesale terms can allocate surplus and discipline downstream participation (Horn and Wolinsky, 1988; Berto Villas-Boas, 2007; Grennan, 2013; Crawford et al., 2018; Collard-Wexler et al., 2019). A related theoretical literature studies how cartel profits are divided using bargaining or profit-sharing rules (Saglam, 2018; Hattori, 2021). Those papers focus on surplus division among heterogeneous firms within a horizontal cartel. In contrast, I study surplus division in a vertically organized cartel. To the best of my knowledge, this is the first paper to model the division of hub-and-spoke cartel profits using a Nash bargaining framework. In my framework, bargaining takes place between the upstream and downstream sides of the market, and the wholesale price determines how the surplus generated by vertical coordination is split between the hub and the spokes.

The findings have direct implications for antitrust enforcement. In competitive settings, vertical coordination may reduce distortions by aligning upstream and downstream incentives. In collusive settings, however, the same instruments can strengthen cartel conduct by weakening fringe competition, inducing broader participation, and transferring surplus through wholesale terms. Treating cartels as a homogeneous category may therefore understate harm when upstream firms actively participate in downstream collusion. Recognizing vertical cartels as a distinct form of collusion is important for measuring damages, designing remedies, and prioritizing enforcement.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting, data, and cartel cases. Section 3 presents the empirical strategy and reduced-form evidence comparing horizontal and vertical cartels. Section 4 develops the theoretical model motivated by the empirical findings. Section 5 presents the calibrated counterfactual exercises. Section 6 concludes.

2 Institutional Background and Data

2.1 Fuel Distribution and Retailing in Brazil

The empirical setting is the Brazilian gasoline retail market, an industry in which retail prices are publicly posted, stations compete in geographically localized markets, and downstream firms maintain observable contractual relationships with upstream fuel distributors. Throughout the paper, I refer to fuel distributors as wholesalers. A central institutional feature of this market is that vertical integration between wholesalers and retailers is legally prohibited: the same firm cannot operate simultaneously as a fuel distributor and a gas station. Vertical relationships therefore arise through contractual arrangements rather than ownership.

The most important contractual arrangement is the exclusive supply agreement. Under these contracts, a gas station agrees to purchase fuel from a single wholesaler and, in exchange, is allowed to operate under the wholesaler’s brand. These agreements structure the relationship between upstream and downstream firms and create a channel through which wholesalers may influence downstream conduct, even in the absence of formal vertical integration. Accordingly, vertical coordination in this industry occurs through exclusivity, branding, and repeated contractual interaction rather than through common ownership.

The geographic organization of firms also matters for cartel formation. Gas stations are typically local operators, often active only within a single city or metropolitan area. By contrast, wholesalers are national firms that supply stations across many regions. Given Brazil’s geographic scale, wholesalers often delegate pricing and commercial decisions to regional managers. This institutional feature is reflected in CADE’s enforcement records: in several vertical cartel cases, penalties are imposed not only on firms but also on regional managers responsible for coordinating conduct in the affected locality. The separation between national ownership and local pricing authority helps explain how vertical collusion can be organized at the market level despite the broader geographic scope of upstream firms.

Gasoline retail markets have been widely studied in the cartel literature because they combine posted prices, frequent interaction, local demand, and relatively homogeneous products. Previous work documents price coordination, cartel stability, and enforcement outcomes in this industry, including detailed analyses of cartel behavior and punishment mechanisms (Clark and Houde, 2013, 2014; Chaves and Duarte, 2025; Byrne and De Roos, 2019). These features make gasoline retail an especially useful setting for comparing horizontal cartels among retailers with vertical cartels involving both retailers and wholesalers.

2.2 Cartel Cases and Legal Records

Cartel investigations in Brazil are conducted by law-enforcement authorities in coordination with the Brazilian Antitrust Authority (CADE). Investigations are based on legally verifiable evidence, such as information from regulatory agencies, leniency agreements, wiretaps, document seizures, or third-party complaints, and aim to document explicit coordination among firms. In this paper, I rely exclusively on publicly available documentation produced by CADE. These records provide detailed information on cartel structure, geographic scope, duration, participating firms, and the role played by upstream and downstream agents.

CADE defines cartel start and end dates based on legal evidence rather than on economic indicators. The start date corresponds to the earliest documented evidence of collusive conduct, while the end date is typically associated with the public disclosure of the investigation, such as arrests, search-and-seizure operations, or the interruption of the documented agreement. For this reason, the treatment window used in the empirical analysis should be interpreted as a legally documented cartel period. It is conservative in the sense that collusive behavior may have begun before the first legally recorded act. At the same time, because treatment dates are not selected using the price series analyzed below, the empirical design avoids mechanically defining cartel periods from the outcomes of interest. The empirical strategy further addresses concerns about differential demand or cost shocks by comparing affected markets to matched nearby control markets and by examining pre-treatment dynamics.

The analysis focuses on eight gasoline retail cartel cases adjudicated by CADE between 2004 and 2020. Four cases involve horizontal coordination among gas stations, while four involve vertical coordination between gas stations and wholesalers. I classify a case as vertical when CADE’s records document active participation by wholesalers or their regional representatives in the coordination of downstream conduct. I classify a case as horizontal when the documented agreement involved coordination among retailers without evidence of upstream participation. Cases are included when firms were convicted or when legal proceedings formally recognized strong cartel evidence, even if final penalties were later affected by procedural issues. Appendix B describes the legal documents underlying each classification.

Table 1 summarizes the main characteristics of each case, including cartel structure, geographic scope, and duration.

Table 1: Cartel Dating

Case Number	Case Legal Code	Cartel Start	Cartel End	Relevant Market	Structure
1	08700.003447/2015-40	Jul/01	May/07	Cuiabá/MT, Várzea Grande/MT	Vertical
2	08012.010215/2007-96	Jul/04	Apr/06	Caxias do Sul/RS	Horizontal
3	08012.007866/2007-07	Jan/06	Apr/07	João Pessoa/PB	Vertical
4	08012.008847/2006-17	Jul/06	Mar/07	Vitória/ES, Cariacica/ES, Vila Velha/ES, Serra/ES	Horizontal
5	08012.011668/2007-30	Jan/07	Aug/07	Cambé/PR, Londrina/PR	Horizontal
6	08700.010769/2014-64	Feb/07	May/08	Belo Horizonte/MG, Betim/MG, Contagem/MG	Vertical
7	08700.000547/2008-95	Sep/08	Dec/09	Teresina/PI	Horizontal
8	08012.008859/2009-86	Jan/11	Feb/16	Brasília/DF	Vertical

Note: This table refers to the dates defined by each legal proceeding. The Legal Code refers to the number attributed by the Antitrust Authority to that cartel case and contains all its documents. Cartel Start and Cartel End refer to the cartel period, while relevant market refers to the cities affected by the cartel.

2.3 Price Data and Sample Construction

Price data come from the weekly Fuel Price Panel collected by the Brazilian Oil, Natural Gas, and Biofuels Agency (ANP). The data report retail gasoline prices charged by individual gas stations and wholesale prices paid by stations to fuel distributors from May 2004 to August 2020. To proxy for wholesalers' marginal costs, I use regional gasoline production and import prices published by ANP. These cost measures vary over time and across regions and capture the main input-cost component faced by wholesalers.

The ANP price survey is not designed as a balanced panel of establishments. As documented by Silveira et al. (2022), stations included in the survey are selected to maximize geographic coverage rather than to follow a fixed firm-specific sampling rule. Consequently, the station-level panel is strongly unbalanced. This sampling design mitigates concerns that inclusion in the price panel is chosen by stations in

response to cartel conduct, although the empirical analysis also relies on station fixed effects and matched control markets to address remaining compositional differences.

I combine the ANP price data with CADE legal records to construct a station-week panel covering multiple cartel episodes within the same industry and institutional environment. Treated stations are those located in cities identified by CADE as affected by cartel conduct during the legally documented cartel period. Potential control stations are located in cities within a 50–200 kilometer radius of the cartel-affected market, following CADE’s market-definition practice. Stations located within a 1–50 kilometer radius are excluded to reduce potential spillover effects from the cartel-affected market.

The final dataset includes 7,644 gas stations across 156 cities. Figure 1 shows the geographic distribution of cartel-affected cities, potential control cities, and matched control cities. The figure also confirms that no cartel-affected city appears in the control group of another case.

Additional station-level characteristics are obtained from the PostoWeb platform. These data include gasoline storage capacity, number of pumps, and exclusive supply agreements with wholesalers. I combine these station-level characteristics with population and vehicle ownership statistics from the Brazilian Institute of Geography and Statistics (IBGE). These variables are used to construct the matching covariates in the empirical analysis. Summary statistics are reported in Table 2.

2.4 Outcome Variables

The main outcomes are retail prices, retail markups, wholesale prices, and wholesale markups. Retail prices are directly observed at the station-week level. Wholesale prices are measured as the price paid by station i to its distributor in week t . Retail markups are defined as the difference between the retail price and the wholesale price, divided by the retail price:

$$\text{Retail Markup}_{it} = \frac{p_{rit} - p_{wit}}{p_{rit}}, \quad (1)$$

where p_{rit} is the retail price charged by station i in week t and p_{wit} is the wholesale price paid by that station.

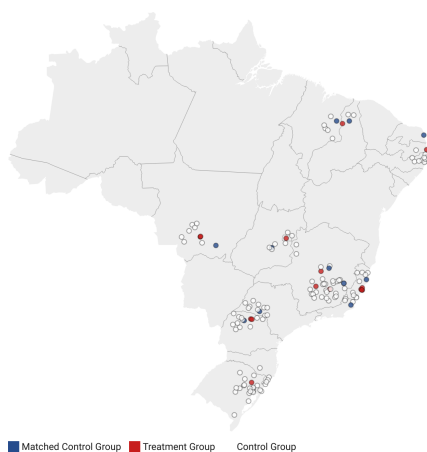
Wholesale markups are defined as the difference between the wholesale price and the regional wholesale cost proxy, divided by the wholesale price:

$$\text{Wholesale Markup}_{it} = \frac{p_{wit} - c_{jt}}{p_{wit}}, \quad (2)$$

where c_{jt} denotes the regional gasoline production and import cost in region j and week t . This measure captures variation in upstream price-cost margins net of observed regional cost movements.

Gasoline demand in Brazil is relatively inelastic. Using household-level data, Santos (2013) estimate a short-run price elasticity of gasoline demand of -0.399 . This low elasticity reinforces the interpretation of retail and wholesale margins as economically meaningful outcomes: changes in margins are informative about how cartel conduct reallocates surplus between consumers, retailers, and wholesalers.

Figure 1: Cartel and Control Cities



Note: The figure plots cartel-affected cities, potential control cities, and matched control cities. Treated stations are located in cities identified by CADE as affected by cartel conduct. Potential control stations are located in cities between 50 and 200 kilometers from the affected market; cities within 50 kilometers are excluded to reduce potential spillover effects.

Table 2: Summary of Statistics

	All Cities	Relevant Market	Matched Cities
Firm-level Data			
Retail Price (BRL)	2.78 (0.592)	2.73 (0.596)	2.83 (0.636)
Wholesale Price (BRL)	2.41 (0.540)	2.41 (0.541)	2.45 (0.573)
Wholesale Cost (BRL)	1.63 (0.340)	1.65 (0.341)	1.66 (0.346)
Retail Markup	0.13 (0.038)	0.12 (.042)	0.13 (0.036)
Wholesale Markup	0.47 (0.088)	0.46 (.087)	0.48 (0.094)
Observations	902,854	261,507	139,960
City-level Data			
HHI retailer (2024)	1,204 (1,486)	317 (240)	418 (411)
HHI wholesaler (2024)	2,023 (1,524)	1,586 (623)	1,329 (724)
Number of Cars (2024)	78,823 (200,037)	362,738 (519,144)	147,292 (180,855)
Population (2024)	213,067 (384,151)	815,219 (799,909)	401,542 (398,884)
Observations	156	15	11

Note: This table presents the summary of statistics. The first column represents the information regarding all firms in the treatment and control group. The second column refers only to the treatment group (firms inside of the relevant market of a cartel). The third column displays the statistics for the firms in the matched control group as defined in Section 3. The standard deviation are in parasyntesis.

3 Documentary Evidence on Cartel Organization

Before turning to the reduced-form evidence, I use the administrative records to characterize how the cartels were organized. The purpose of this section is not to re-litigate liability, but to document the mechanisms through which the cartels operated in the cases studied in the paper. The comparison reveals a central distinction. Horizontal cartels relied primarily on direct coordination among retailers. Vertical cartels expanded the cartel’s organizational technology by incorporating wholesalers into monitoring, enforcement, and the allocation of margins along the supply chain.

In the horizontal cases, coordination was organized at the retail level. Retailers used meetings, phone calls, trade associations, reference stations, and local leaders to agree on prices and monitor deviations. In Teresina/PI (Case 7), the administrative record indicates that the retail fuel association was used as a forum for coordinating commercial conduct. Meeting minutes instructed station owners to avoid “price wars and payment-term disputes,” and the decision describes later episodes of price competition as “price wars” that were followed by efforts to restore margins. In one statement, a representative described that, after prices fell, retailers concluded that “the category would go bankrupt,” after which one of them took the initiative to return prices to the earlier level. The language used in the records is informative: price competition itself was treated as a breakdown of coordination.

The Vitória/ES cartel (Case 4) provides direct evidence of horizontal coordination among competing retailers. The administrative record describes repeated communications between gas station owners about the retail prices to be charged at specific stations and in specific local areas. Because these conversations concerned illegal price coordination, participants often avoided explicit references to gasoline prices. Instead, they used coded expressions such as “math,” “engine,” and “machine configuration” to refer to price levels or price changes. The relevant point is not the literal meaning of these expressions, but the mechanism they reveal: retailers communicated directly with one another about the prices that would be posted at the pump, checked whether other stations were following the coordinated price, and adjusted their own prices in response to the behavior of rival retailers. This evidence therefore illustrates the basic organization of a horizontal retail cartel. Coordination and monitoring occurred among competing stations themselves, without distributors serving as active enforcers of the agreement.

The horizontal cases also show that direct coordination did not require the participation of every station in the market. In Caxias do Sul/RS (Case 2) and Londrina/PR (Case 5), the records describe leader-follower structures in which a small number of large or strategically located stations served as focal points for price coor-

dination. Once these reference stations adjusted their prices, smaller stations tended to follow. In Londrina/PR, for example, an intercepted call describes simultaneous price changes and indicates that removing the old number from the price sign was interpreted as acceptance of the coordinated adjustment. This organizational structure is similar to the mechanism documented in other gasoline retail cartels, such as Clark and Houde (2014) and Byrne and De Roos (2019), where a leading station initiates price changes and a set of followers subsequently adjusts their own prices.

The Brazilian records also show that horizontal cartels relied on active enforcement within the retail market. In addition to leader-follower price coordination, the documents describe subgroups of retailers responsible for visiting and pressuring stations that had not yet aligned with the agreement. Thus, horizontal cartels reduced communication costs by relying on local leaders, visible posted prices, and reference stations. However, enforcement remained largely downstream: retailers monitored pump prices, exchanged phone calls, held meetings, applied peer pressure, and used direct threats to discipline deviations.

Vertical cartels exhibit a different enforcement technology. In these cases, the relevant distinction is not only whether a distributor firm was formally condemned, but whether agents operating at the wholesale level participated in the organization, monitoring, or enforcement of the retail agreement. The administrative records often distinguish between the liability of the distributor as a legal entity and the conduct of its local employees, managers, or commercial representatives. This distinction matters for the economic mechanism. Local wholesaler representatives are the agents who interact repeatedly with retailers, observe supply relationships, transmit information across stations, and can influence discounts, credit, delivery conditions, or other wholesale terms. For this reason, I classify a cartel as involving vertical participation when the records document the active involvement of wholesalers or their representatives, even when the legal conclusion concerning the distributor firm itself is weaker, archived, or treated separately from the conduct of individual representatives.

The first mechanism is that wholesalers can expand the cartel’s communication network. A horizontal retail cartel requires station owners to coordinate directly with one another, often through meetings, phone calls, or a retail association. When wholesalers or their representatives participate, the cartel gains an additional channel through which information can circulate across retailers. The João Pessoa/PB record illustrates this mechanism. The technical note describes the retail association as the main organizer of downstream coordination, but it also states that the distributor Ello-Puma appeared to serve as a forum for discussions related to the alleged price cartel. The records further indicate that distributor representatives were informed

about whether the agreement was being maintained or broken and that the retail association sought to mobilize the distributor in response to stations that were not aligned. Thus, the vertical channel did not replace horizontal coordination among retailers; instead, it complemented it by providing an additional institutional and commercial link through which the agreement could be monitored and sustained.

The second mechanism is enforcement through pressure on non-compliant retailers. Horizontal cartels can discipline deviations through peer pressure, visits, threats, and future retail price retaliation. Vertical participation adds a different instrument: the wholesaler has an ongoing commercial relationship with the station. This relationship can be used to pressure retailers that deviate from the agreed price or that remain outside the cartel. The Belo Horizonte/MG record provides the clearest example. The decision describes wholesalers and their representatives as helping to guarantee coordinated price adjustments by monitoring prices and intervening with stations that did not comply. The central point is that enforcement no longer depends only on rival retailers observing pump prices and applying pressure. Once upstream agents are involved, the cartel can also use brand relationships, supply links, and commercial dependence to discipline retailers.

The third mechanism is enforcement through wholesale terms. The Belo Horizonte/MG documents state that wholesalers offered financial offsets in the amounts charged to retailers when retail prices were changed according to their instructions. The decision also states that stations would receive or stop receiving these financial offsets depending on their behavior. This is the documentary counterpart of the model's key mechanism. In a purely horizontal cartel, a retailer that cuts price faces punishment from other retailers. In a vertical cartel, a retailer that deviates may also lose favorable wholesale conditions, receive lower discounts, or face a higher effective input price. Wholesaler participation therefore makes deviation less attractive by directly affecting the retailer's outside option.

The fourth mechanism is rent allocation along the vertical chain. In a horizontal retail cartel, retailers coordinate the final consumer price but take wholesale terms as given. In a vertical cartel, the cartel can coordinate both the final retail price and the wholesale terms that determine how surplus is divided between wholesalers and retailers. The Cuiabá/MT record is useful for this point because it documents discussions over margins and the division of the retail margin between distribution and retail. This evidence suggests that vertical participation can transform a downstream price-fixing arrangement into a broader mechanism for allocating cartel rents along the supply chain.

The Brasília/DF case reinforces the need to distinguish the economic mecha-

nism from the final legal treatment of each defendant. The retail cartel is strongly documented in the records, while the distributor-related allegations were archived for insufficient evidence. Nevertheless, the case file shows that the vertical allegations concerned precisely the mechanisms relevant to this paper: discrimination among buyers, influence toward uniform commercial conduct, and market division. Therefore, I treat Brasília/DF cautiously. It is not used as the strongest evidence of distributor liability, but it is informative about the type of vertical conduct that investigators associated with the operation of retail fuel cartels.

Table 3 summarizes the documented mechanisms for each cartel case, while Appendix B provides the case-specific evidence from the administrative records. This documentary evidence motivates the empirical comparison. Horizontal cartels were able to coordinate retail prices, monitor posted prices, and pressure retailers that deviated from the agreement. Vertical cartels, however, had access to an additional enforcement channel. By incorporating wholesalers or their local representatives, these cartels could extend monitoring to the wholesale relationship, use wholesalers' commercial networks to reach a larger set of stations, and discipline non-compliance through wholesale prices, discounts, financial offsets, or other contractual terms. The reduced-form analysis therefore asks whether cartels with this vertical enforcement technology generate larger effects on retail prices, retail margins, and wholesale margins than purely horizontal retail cartels. After presenting the empirical results, I develop a model that formalizes this mechanism by allowing wholesalers to participate in the cartel, support a more inclusive downstream agreement, and reduce the attractiveness of non-compliance by worsening the wholesale terms faced by retailers outside the agreement.

4 Horizontal vs. Vertical Cartels

This section compares the effects of horizontal and vertical cartel structures on retail prices, retail markups, and wholesale markups. The central empirical question is whether vertical coordination—that is, coordination involving both retailers and wholesalers—generates different market outcomes from horizontal coordination among retailers alone. The answer is important because vertical participation may change both the distribution of cartel surplus across the supply chain and the ability of retailers to sustain coordinated pricing.

I proceed in two steps. First, I estimate city-level effects of cartel dissolution and test whether these effects differ between horizontal and vertical cartel cases. Second, I use station-level matched panels to examine whether the effects vary with firm

Table 3: Documented Cartel Mechanisms by Case

Case	Type	Participants	Communication	Assoc.	Leader	Retail mon.	Retail pres.	Whol. reps.	Whol. pres.	Whol. terms
Case 1	V	Retailers, retail association, wholesaler representatives	Association meetings and direct contacts among retailers and upstream agents	Yes	Partial	Yes	Yes	Yes	Partial	Yes
Case 2	H	Retailers and large retail networks	Phone calls, meetings, and direct coordination among retailers	No	Yes	Yes	Yes	No	No	No
Case 3	V	Retailers, retail association, wholesaler representatives	Association meetings, phone calls, and contacts involving upstream agents	Yes	Partial	Yes	Yes	Yes	Partial	Partial
Case 4	H	Retailers	Phone calls, coded language, and meetings	No	No	Yes	Partial	No	No	No
Case 5	H	Retailers and reference stations	Phone calls and coordinated timing of price changes	No	Yes	Yes	Yes	No	No	No
Case 6	V	Retailers, retail association, reference stations, wholesaler representatives	Association contacts, direct calls, and vertical coordination between retailers and upstream agents	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Case 7	H	Retailers and retail association	Union meetings, meeting minutes, and direct statements	Yes	No	Yes	Yes	No	No	No
Case 8	V	Retailers, large networks, wholesaler representatives	Direct contacts, network coordination, and intercepted communications	Partial	Yes	Yes	Yes	Partial	Partial	Partial

Notes: H denotes horizontal cartels and V denotes cartels with vertical participation. “Assoc.” indicates participation of a retail association or union in the organization of the cartel. “Leader” indicates the presence of leader firms or reference stations. “Retail mon.” indicates monitoring of posted retail prices. “Retail pres.” indicates pressure, threats, visits, or other downstream enforcement against deviating retailers. “Whol. reps.” indicates participation, accusation, or documented involvement of wholesalers’ local employees, managers, or commercial representatives. “Whol. pres.” indicates pressure through the wholesale relationship, brand relationship, or supply network. “Whol. terms” indicates evidence that upstream agents could influence discounts, effective wholesale prices, financial offsets, or the division of margins between wholesale and retail. “Partial” indicates that the mechanism appears in the record but is weaker, indirect, alleged, archived, or not central to the final legal characterization of the case.

characteristics. The firm-level analysis focuses on station size, which is the mechanism developed in the theoretical model below. Additional heterogeneity results by exclusive supply agreements, together with robustness checks for the matching procedure and event windows, are reported in Appendix C.

4.1 Empirical Design

The empirical strategy compares cartel-affected markets to nearby non-cartelized markets that are similar in observable market characteristics. I first match each cartel-affected city to a control city using retailer concentration, wholesaler concentration, population, and the number of registered vehicles. Because historical concentration measures are not consistently available, I use the most recent available information, measured in 2024. The matching variables are standardized and matched using nearest-neighbor Euclidean distance, with replacement. This procedure selects comparison cities that are similar in market size, potential demand, and supply-side concentration at both levels of the vertical chain. Appendix C.1 provides additional details on the city-level matching procedure and reports the resulting matches.

I then construct a station-level matched sample. Within each treated city and its matched control city, I use Coarsened Exact Matching to match treated stations to comparable control stations observed in the same calendar week. Stations are matched on gasoline storage capacity, number of pumps, and exclusive supply status. The resulting matched panel contains 53,948 matched treated station-week observations out of 74,048 treated station-week observations, corresponding to a matching rate of approximately 73 percent. Appendix C.2 describes the station-level matching procedure in detail.

The main empirical event is cartel dissolution. I focus on dissolution rather than cartel formation because the law-enforcement intervention that terminates the cartel is more precisely observed. CADE identifies the beginning of a cartel using the earliest available legal evidence, but effective coordination may have begun before that date. By contrast, dissolution is typically associated with arrests, search-and-seizure operations, or public disclosure of the investigation, providing a more consistent event date across cases.

The interpretation of the estimates is therefore reversed relative to a standard treatment-onset design. The pre-dissolution period corresponds to the period in which the cartel was operating, while the post-dissolution period corresponds to the period after cartel disruption. Thus, a decline in prices or markups after dissolution implies that the cartel had sustained higher prices or markups while active.

4.2 City-Level Effects

I begin with an event-study analysis at the city level. I estimate

$$Y_{mct} = \sum_{k=-10}^{-2} \lambda_k D_{mct}^k + \sum_{k=0}^{20} \lambda_k D_{mct}^k + \psi_{c,t} + \alpha_m + \varepsilon_{mct}, \quad (3)$$

where Y_{mct} denotes the outcome for city m , cartel case c , and week t . The outcomes are the average log retail price, average retail markup, and average wholesale markup. The indicator D_{mct}^k equals one if city m belongs to the cartel-affected market in case c and week t is k weeks relative to cartel dissolution. The specification includes city fixed effects, α_m , and case-by-calendar-week fixed effects, $\psi_{c,t}$, which absorb shocks common to the treated city and its matched control in each cartel case. The omitted event week is $k = -1$, the week immediately preceding dissolution. Standard errors are clustered at the city level.

Figure 2 reports the event-study estimates. The coefficients in the weeks preceding dissolution are close to zero across the three outcomes. Because the cartel is already operating during this period, these estimates are not conventional pre-treatment tests for cartel formation. Instead, they show that treated and matched control cities did not exhibit differential changes immediately before the law-enforcement intervention. This reduces concerns that the post-dissolution estimates are driven by trends already underway before dissolution.

Following cartel dissolution, retail prices and retail markups decline relative to matched control cities. Because dissolution marks the end of cartel activity, these negative post-dissolution coefficients imply that cartel operation sustained higher retail prices and higher retail markups. Wholesale markups do not exhibit a statistically significant change in the pooled event-study specification, but this pooled estimate masks heterogeneity between horizontal and vertical cartel structures.

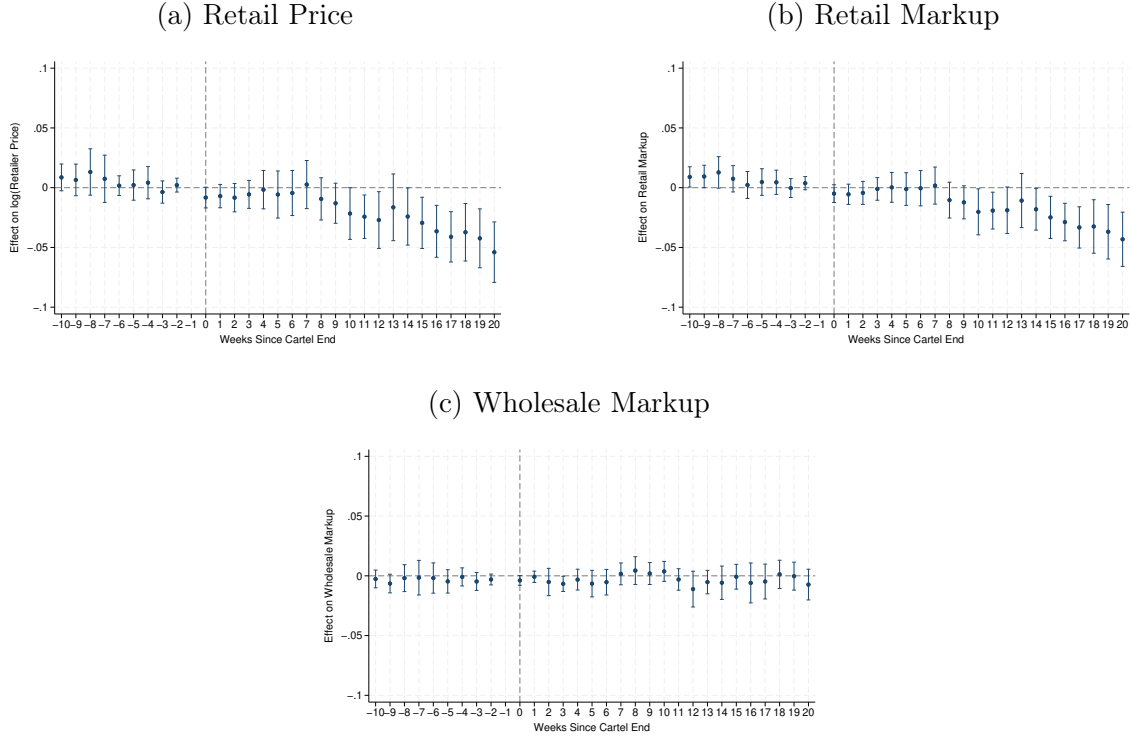
To estimate the average effects of horizontal and vertical cartels, I next use a symmetric window around cartel dissolution. Because cartel episodes differ substantially in duration, I cap the pre-dissolution period at 78 weeks, corresponding to approximately 18 months. For cartel episodes lasting fewer than 78 weeks, I retain the full documented cartel period. For each case, I then use a post-dissolution comparison window of the same length as the retained pre-dissolution period.

Using this matched city-level panel, I estimate

$$Y_{mct} = \alpha_m + \psi_{c,t} + \beta_1 \text{PostCartel}_{mct} + \beta_2 (\text{Vertical}_c \times \text{PostCartel}_{mct}) + \xi_{mct}, \quad (4)$$

where PostCartel_{mct} equals one for the treated city in case c during the post-dissolution period and zero otherwise. The indicator Vertical_c equals one when the case involved

Figure 2: City-Level Event-Study Estimates



Note: Dots report estimated event-time coefficients from specification (3). The omitted category is the week immediately preceding cartel dissolution, $k = -1$. Vertical bars represent 95% confidence intervals. Time 0 corresponds to the week of cartel dissolution. Observations are weighted to account for the frequency with which a control city is selected in the city-level matching procedure. Standard errors are clustered at the city level.

vertical coordination between retailers and wholesalers. The specification includes city fixed effects and case-by-calendar-week fixed effects. Standard errors are clustered at the city level.

Because the omitted period is the cartel period, β_1 measures the post-dissolution change for horizontal cartels. A negative estimate of β_1 therefore implies that horizontal cartels increased the outcome while active. Similarly, β_2 measures the additional post-dissolution change for vertical cartels relative to horizontal cartels. A negative estimate of β_2 implies that vertical cartels generated larger increases in the outcome during the cartel period.

Table 4 reports the estimates. Retail prices and retail markups decline after the

Table 4: City-Level: Effect of Cartel Dissolution

	(1)	(2)	(3)
	log(Retail Price)	Retail Markup	Wholesale Markup
Post_Cartel	-0.012*** (0.004)	-0.011*** (0.002)	-0.000 (0.002)
Post_Cartel $\times$ Vertical	-0.032*** (0.010)	-0.015** (0.007)	-0.021* (0.011)
Constant	0.986*** (0.001)	0.143*** (0.001)	0.459*** (0.000)
Case-Week Fixed Effects	Yes	Yes	Yes
City Fixed Effects	Yes	Yes	Yes
Observations	3,399	3,399	3,399

Notes: This table reports estimates from specification (??). All regressions include case-by-calendar-week fixed effects and city fixed effects. Standard errors are clustered at the city level and are reported in parentheses. The omitted period corresponds to the period in which the cartel was operating; therefore, negative post-dissolution coefficients imply higher outcomes during cartel operation. * (p<0.10), ** (p<0.05), *** (p<0.01).

dissolution of horizontal cartels, implying that horizontal cartel activity increased both outcomes during the cartel period. The interaction between PostCartel and Vertical is also negative and statistically significant for both outcomes. The estimates imply that horizontal cartels increased retail prices by approximately 1.4 percent, whereas vertical cartels increased retail prices by approximately 4.4 percent. Thus, the retail-price effect of vertical collusion is roughly three times as large as the effect of horizontal collusion.

Retail markups display a similar pattern. Horizontal cartels increased retail markups by approximately 1.1 percentage points, while vertical cartels generated an additional increase of approximately 2.6 percentage points. These estimates indicate that the participation of wholesalers substantially increased the gains accruing to retailers.

The evidence for wholesale markups is weaker but consistent with the same interpretation. The post-dissolution effect for horizontal cartels is close to zero and

statistically insignificant, as expected if wholesalers were not participants in horizontal retail cartels. By contrast, the interaction between PostCartel and Vertical is negative and statistically significant at the 10 percent level. This implies that wholesale markups declined after the dissolution of vertical cartels, consistent with wholesalers benefiting from vertical coordination while the cartel was active.

Because the comparison between horizontal and vertical cartels is based on only eight cases, I supplement conventional inference with an exact randomization-inference exercise. With eight cartel cases and four vertical cases, there are $\binom{8}{4} = 70$ possible assignments of four cases to the vertical-cartel category. I re-estimate specification (4) under each assignment and compare the interaction coefficient under the actual classification with the resulting placebo distribution. The actual interaction coefficient ranks as the third most negative estimate for log retail prices, the sixth most negative estimate for retail markups, and the fourteenth most negative estimate for wholesale markups. These rankings provide strong support for the retail-price result, suggestive support for the retail-markup result, and weaker evidence for the wholesale-markup result.

Overall, the city-level analysis shows that vertical cartels generate larger retail-market effects than horizontal cartels. The evidence for wholesale markups is less precise, but the estimates are consistent with wholesalers benefiting only when they participate directly in the collusive arrangement. Appendix C shows that these patterns are robust to alternative event windows, monthly aggregation, and excluding weeks immediately surrounding the law-enforcement intervention.

4.3 Firm-Level Heterogeneity by Station Size

The city-level estimates establish that vertical cartels have larger effects than horizontal cartels, but they do not explain why. I therefore turn to station-level evidence. The theoretical model developed below emphasizes the role of small retailers. In horizontal cartels, small retailers may be harder to discipline because they internalize a smaller share of the gains from coordinated pricing and may have stronger incentives to undercut the cartel. Vertical coordination can alter these incentives by changing wholesale terms and by allowing upstream firms to help discipline downstream conduct.

Using the matched station-level panel, I estimate stacked difference-in-differences specifications of the form

$$Y_{ist} = \alpha_{is} + \lambda_{st} + \beta_1 D_{ist} + \beta_2 (D_{ist} \times \text{Vertical}_s) + \beta_3 (D_{ist} \times X_i) + \beta_4 (D_{ist} \times \text{Vertical}_s \times X_i) + \varepsilon_{ist}, \quad (5)$$

where Y_{ist} denotes the outcome of station i in matched stack s and week t . Each stack contains one treated station and its matched control stations. The variable

$$D_{ist} = \text{Treated}_{is} \times \text{Post}_{st}$$

equals one for treated stations after cartel dissolution. The specification includes station-by-stack fixed effects, α_{is} , and stack-by-week fixed effects, λ_{st} . Identification therefore comes from comparing the change in outcomes for a treated station after cartel dissolution with the contemporaneous change among observationally similar control stations in the same matched stack.

I define small firms at the level of the gas-station ownership group. Specifically, within each treated market, I aggregate storage capacity across all stations belonging to the same CNPJ root and classify as small the groups within the first quartile of the total storage capacity distribution. This definition captures the market position of the retail firm rather than the capacity of an individual establishment.

Table 5 illustrates the result for these regressions. Panel A reports total effects by cartel type and station size, while Panel B decomposes these totals into the individual regression components. Starting with retail prices, Panel A shows positive effects for all combinations of cartel organization and station size. Retail prices increase by 2.2% for large stations and 2.1% for small stations in horizontal cartels, and by 3.8% for large stations and 3.3% for small stations in vertical cartels. This indicates that retail prices were higher during cartel operation for both large and small stations, regardless of whether the cartel was horizontal or vertical. Consistent with the city-level results, the effects are larger in vertical cartels than in horizontal cartels. Panel B reinforces this interpretation: the additional vertical-cartel component for large stations is 1.7% and statistically significant, implying that vertical organization is associated with a larger increase in retail prices. At the same time, the table provides little evidence that the retail-price effect differs systematically by station size. The small-station component in horizontal cartels and in the vertical cartels are close and statistically insignificant.

The results for retail markups show a similar pattern in terms of the overall cartel effect, but with an important difference across station size. Panel A shows that all retail-markup effects are positive: retail markups increase by 1.9% for large stations and 1.7% for small stations in horizontal cartels, and by 3.2% for large stations and 2.0% for small stations in vertical cartels. These estimates indicate that cartels were able to raise downstream margins during the period of operation. As in the city-level analysis, vertical cartels generate larger retail-markup effects than horizontal cartels for large stations. The additional vertical-cartel component for large stations is 1.3% and statistically significant. However, the retail-markup gain for small stations in

Table 5: Firm-Level Effects by Station Size

	(1) Retail Price	(2) Retail Markup	(3) Wholesale Markup
<i>Panel A. Total effects by cartel type and station size</i>			
Horizontal cartel, large station	0.0215*** (0.0019)	0.0187*** (0.0020)	-0.0001 (0.0011)
Horizontal cartel, small station	0.0206*** (0.0025)	0.0166*** (0.0025)	0.0010 (0.0011)
Vertical cartel, large station	0.0381*** (0.0024)	0.0319*** (0.0026)	0.0002 (0.0016)
Vertical cartel, small station	0.0327*** (0.0043)	0.0204*** (0.0026)	0.0062*** (0.0021)
<i>Panel B. Individual components</i>			
Horizontal cartel effect, large stations	0.0215*** (0.0019)	0.0187*** (0.0020)	-0.0001 (0.0011)
Small-station component in horizontal cartels	-0.0009 (0.0031)	-0.0020 (0.0031)	0.0011 (0.0015)
Additional vertical-cartel component, large stations	0.0167*** (0.0030)	0.0132*** (0.0033)	0.0003 (0.0019)
Additional small-station component in vertical cartels	-0.0045 (0.0056)	-0.0094** (0.0048)	0.0049* (0.0026)
Stack-Firm Fixed Effects	Yes	Yes	Yes
Stack-Week Fixed Effects	Yes	Yes	Yes
Observations	96,442	96,442	96,442
Number of Stations	1,138	1,138	1,138

Notes: This table reports estimates from stacked difference-in-differences regressions estimated on the matched firm-level sample. Because treatment is defined as the period after cartel dissolution, reported coefficients reverse the sign of the post-dissolution estimates and are interpreted as implied effects during cartel operation. Panel A reports total effects for each cartel type and station-size group. Panel B reports the individual contribution of each regression component to those total effects. A station is classified as small according to the definition described in the data section. All specifications include stack-by-firm and stack-by-week fixed effects and are estimated using matching weights. Standard errors clustered at the station level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

vertical cartels is lower than the gain for large stations, where small firms have 1% lower benefit than the big firms under Vertical cartel. This pattern is consistent with the mechanism documented in Section 2. If wholesalers use wholesale prices to discipline or punish smaller retailers that are more likely to deviate, then part of the cartel gain is transferred upstream through higher wholesale charges, reducing the retail-margin gains of these stations.

The wholesale-markup results support this interpretation. In Panel A, wholesale markups do not increase meaningfully for large stations in either horizontal or vertical cartels, the estimated effects for both horizontal and vertical cartels are virtually zero. The effect for small stations in horizontal cartels is also small, at 0.0010 and statistically insignificant. By contrast, the wholesale-markup effect for small stations in vertical cartels is 0.6% and statistically significant. Panel B shows that the additional small-station component in vertical cartels is positive, at 0.5%. Therefore, the evidence suggests that the vertical organization of the cartel did not simply raise retail prices uniformly across firms. Instead, it appears to have redistributed part of the surplus away from small downstream firms and toward wholesalers, consistent with a vertical enforcement mechanism in which wholesalers charge higher wholesale margins to smaller stations.

5 Theoretical Model

The documentary evidence shows that horizontal and vertical cartels differ in their internal organization. In horizontal cartels, the legal records frequently describe discussions about retailers that deviate from the agreement by charging prices below the collusive level. In vertical cartels, by contrast, two additional mechanisms appear. First, wholesalers participate directly in discussions over retail prices and retail markups. Second, wholesalers use their vertical relationship with retailers to support the agreement: they provide more favorable terms to stations that participate in the cartel and penalize retailers that deviate by charging higher wholesale prices or offering less favorable contracts.

To connect this evidence on cartel organization with the reduced-form results, I develop a model of cartel formation in a vertically related market. The purpose of the model is to formalize the mechanisms documented in the legal records and show how they can generate the empirical patterns found in the data. In the model, horizontal retail cartels are constrained by deviating retailers and by the presence of a competitive fringe. When wholesalers participate, however, they can use wholesale prices to weaken the outside option of non-participating retailers, sustain broader

downstream participation, and divide the additional surplus generated by the cartel.

The model is close in spirit to Clark et al. (2024), who distinguish between different cartel structures depending on whether collusion occurs upstream, downstream, or vertically across the supply chain. I adopt a similar classification of cartel cases, but adapt the framework to a capacity-constrained downstream retail market following Bos and Harrington (2010). This allows the model to capture how retail cartel participation depends on firm capacity and how the size of the downstream cartel affects the sustainable retail price.

The main departure from the standard horizontal-cartel framework is that wholesalers can affect the profitability and stability of downstream collusion through the wholesale price. When wholesalers and retailers do not coordinate, the wholesale markup changes the retail cartel’s effective marginal cost and may generate double marginalization. When wholesalers and retailers coordinate vertically, however, wholesalers can facilitate downstream collusion by discriminating between cartel participants and non-participants through wholesale prices. This mechanism is related to Van Cayseele and Miegielsen (2013), who study how an upstream firm can support downstream collusion by affecting retailers’ outside options. In my setting, wholesalers do not necessarily exclude non-participants from the market, but they can make non-participation less attractive by charging outsiders a higher wholesale price.

5.1 Setting

The economy consists of two vertically related markets: an upstream wholesale market and a downstream retail market. A homogeneous good is sold to final consumers, who have demand $D(p)$. I assume that $D(p)$ is twice continuously differentiable, strictly decreasing, and weakly concave in the retail price p .

The upstream market consists of N_w homogeneous wholesalers, each with constant marginal cost c and discount factor $\delta \in (0, 1)$. The downstream market consists of N_r retailers with discount factor $\theta \in (0, 1)$. Each retailer i has storage capacity k_i , and total retail capacity is

$$K = \sum_{i=1}^{N_r} k_i.$$

Retailers purchase the good at wholesale price p_w and sell it to final consumers at retail price p_r . Thus, the wholesale price is the marginal cost of downstream firms.

As in Bos and Harrington (2010), capacity constraints play a central role. I impose two assumptions on retail capacities.

Assumption 1. Let $p^M(p_w)$ denote the monopoly retail price given wholesale price p_w :

$$p^M(p_w) \in \arg \max_p (p - p_w)D(p).$$

For every retailer i ,

$$k_i < D(p^M(p_w)).$$

Thus, no single retailer can serve monopoly demand.

Assumption 2. For every retailer i ,

$$\sum_{j \neq i} k_j \geq D(p_w).$$

Thus, if any one retailer is removed from the market, the remaining retailers can jointly serve competitive demand.

These assumptions capture the idea that gasoline stations face physical storage limits. A single station cannot supply the entire local market, but the market remains competitive if one station is removed. The model abstracts from additional retail marginal costs and from horizontal differentiation across gas stations. Costs such as labor, rent, and station operation are treated as fixed. Retailers are assumed to sell a homogeneous product and face a common market demand. These simplifications allow the analysis to focus on the interaction between wholesale pricing, retail capacity, and cartel stability.

Timing. The model has two stages. In the first stage, firms determine the cartel structure. In the second stage, prices are chosen in an infinitely repeated pricing game. The timing is as follows.

1. Cartel-formation stage.

- (a) Retailers decide whether to form a downstream cartel. If a downstream cartel is formed, retailers choose the set of cartel members Γ and the aggregate cartel capacity

$$K_\Gamma = \sum_{i \in \Gamma} k_i.$$

- (b) Wholesalers decide whether to form an upstream cartel.

- (c) If both an upstream cartel and a downstream cartel are formed, the two cartels decide whether to coordinate vertically. If they coordinate, the market is characterized by a vertical cartel. If they do not coordinate, the upstream and downstream cartels choose prices independently.
2. **Repeated pricing stage.** After the cartel structure is determined, firms play an infinitely repeated pricing game. The pricing game depends on the cartel structure chosen in the first stage.
- (a) **No vertical coordination.** If there is no vertical coordination between wholesalers and retailers, prices are chosen sequentially. First, wholesalers choose the wholesale price p_w . Then, taking p_w as given, retailers choose the retail price p_r . This case includes markets with no cartel, only an upstream cartel, only a downstream cartel, or both cartels operating independently.
- (b) **Vertical coordination.** If the upstream and downstream cartels coordinate, wholesale and retail prices are jointly determined through Nash bargaining between the two cartels. The bargaining outcome specifies the wholesale price and the retail price.

This timing nests five market structures:

- **Non-cartel benchmark:** wholesalers compete à la Cournot and retailers compete à la Bertrand.
- **Wholesale cartel:** wholesalers form an upstream cartel while retailers remain competitive.
- **Retail cartel:** retailers form a capacity-constrained downstream cartel while wholesalers remain non-cooperative.
- **Double horizontal cartel:** wholesalers and retailers both form cartels, but they do not coordinate vertically.
- **Vertical cartel:** wholesalers and retailers coordinate vertically and jointly determine wholesale and retail prices.

I solve the model by backward induction. First, in the next five subsections, I characterize the pricing outcome under each market structure. Then, in Subsection 4.7, I derive the conditions under which each outcome can be sustained in equilibrium.

5.2 Non-Cartel Benchmark

I begin with the non-cartel benchmark. Wholesalers compete à la Cournot, so the wholesale price includes a Cournot markup over marginal cost:

$$p_w^N = c + \mu_w^N,$$

where $\mu_w^N \geq 0$ is decreasing in the number of wholesalers N_w .

Retailers compete in prices. Since the product is homogeneous and all retailers face the same wholesale price, Bertrand competition implies that retailers price at marginal cost:

$$p_r^N = p_w^N.$$

Thus, retailers earn zero variable profits, while wholesalers may earn positive profits due to upstream market power. This benchmark serves as the reference point for the cartel cases below.

5.3 Wholesale Cartel

I next consider the case in which wholesalers form an upstream cartel while retailers remain competitive. Since retailers compete à la Bertrand, the wholesale price is passed through to final consumers:

$$p_r = p_w.$$

The wholesale cartel therefore chooses the wholesale price to maximize joint upstream profits,

$$p_w^{WC} \in \arg \max_{p_w} (p_w - c)D(p_w),$$

and the retail price is

$$p_r^{WC} = p_w^{WC}.$$

Total wholesale-cartel profit is

$$\Pi_w^{WC} = (p_w^{WC} - c)D(p_w^{WC}),$$

while retailers earn zero variable profits. With symmetric sharing, each wholesaler receives Π_w^{WC}/N_w .

The wholesale cartel is sustained by grim-trigger strategies. If one wholesaler deviates, it slightly undercuts the cartel price and captures the entire wholesale market

for one period. After a deviation, wholesalers revert forever to Bertrand competition, which drives the wholesale price to marginal cost and yields zero continuation profits. The incentive constraint is therefore

$$\frac{1}{1-\delta} \frac{\Pi_w^{WC}}{N_w} \geq \Pi_w^{WC}.$$

Equivalently,

$$\delta \geq 1 - \frac{1}{N_w}. \quad (6)$$

Under this punishment structure, the sustainability of upstream collusion depends only on the number of wholesalers. Demand affects the level of wholesale profits, but both the collusive payoff and the one-period deviation payoff are proportional to Π_w^{WC} , so the incentive constraint depends only on N_w .

5.4 Retail Cartel

I now consider the case in which retailers form a downstream cartel while wholesalers remain non-cooperative. Retailers take the wholesale price as given and decide whether to participate in a capacity-constrained cartel, as in Bos and Harrington (2010).

Let $\Gamma \subseteq \{1, \dots, N_r\}$ denote the set of cartel members, and let

$$K_\Gamma = \sum_{i \in \Gamma} k_i$$

be their aggregate capacity. Cartel profits are shared proportionally to capacity, so that member $i \in \Gamma$ receives

$$\pi_i^{RC} = \frac{k_i}{K_\Gamma} \Pi_\Gamma^{RC}.$$

This sharing rule gives larger retailers a larger share of cartel surplus and therefore stronger incentives to participate.

Retailers outside the cartel form a competitive fringe. They slightly undercut the cartel price and sell up to capacity. Thus, while cartel members charge p_r , outsiders charge $p_r - \xi$, for some $\xi > 0$. As in Bos and Harrington (2010), I consider the limit as $\xi \rightarrow 0$. In this case, an outside retailer i sells its full capacity and earns

$$(p_r - p_w)k_i.$$

The retail cartel, in turn, serves residual demand after fringe capacity is supplied:

$$D(p_r) - (K - K_\Gamma),$$

where $K - K_\Gamma$ is the total capacity of retailers outside the cartel.

The Bos–Harrington incentive constraint requires each cartel member to prefer remaining in the cartel rather than deviating, undercutting the cartel price, and then reverting to the non-collusive punishment path. If cartel profits are shared in proportion to capacity, retailer i 's per-period cartel profit is

$$[D(p_r) - (K - K_\Gamma)] (p_r - p_w) \frac{k_i}{K_\Gamma}.$$

The no-deviation constraint is therefore

$$\frac{1}{1 - \theta} [D(p_r) - (K - K_\Gamma)] (p_r - p_w) \frac{k_i}{K_\Gamma} \geq (p_r - p_w) k_i.$$

After canceling common terms, this condition becomes

$$D(p_r) \geq K - \theta K_\Gamma. \tag{7}$$

Because demand is decreasing, equation (7) places an upper bound on the retail price that the cartel can sustain. A larger cartel capacity K_Γ relaxes the constraint and allows the cartel to sustain a weakly higher price.

This structure has two important implications for cartel composition. First, the retail cartel is generally not fully inclusive. In equilibrium, there is a cutoff capacity $\bar{k}$ such that retailer i joins the cartel if and only if

$$k_i \geq \bar{k}.$$

Thus, the cartel is formed by the largest retailers, while smaller retailers remain in the competitive fringe. Second, conditional on cartel formation, cartel members prefer the cartel to be larger: additional cartel capacity relaxes (7) and raises the maximum sustainable cartel price. These results follow from Bos and Harrington (2010) and are important for the mechanism below. They imply that horizontal retail cartels are constrained by the participation and discipline of small retailers, which may remain outside the cartel and limit the price that can be sustained.

Given K_Γ , the downstream cartel chooses the retail price that maximizes residual-demand profits subject to (7). The pricing rule is

$$p_r^{RC}(p_w) = \min\{p^*(p_w, K_\Gamma), \bar{p}(K_\Gamma)\}, \tag{8}$$

where $p^*(p_w, K_\Gamma)$ is the unconstrained residual-monopoly price and

$$\bar{p}(K_\Gamma) = D^{-1}(K - \theta K_\Gamma)$$

is the highest sustainable price. The unconstrained residual-monopoly price satisfies

$$D(p^*) - (K - K_\Gamma) + (p^* - p_w)D'(p^*) = 0. \quad (9)$$

The retail-cartel pricing game captures several features of the legal evidence discussed in Section 2. The documents frequently describe retailers charging prices below the collusive level and cartel members attempting to pressure or intimidate them into aligning with the agreement. In the model, these firms are represented by the competitive fringe: retailers that remain outside the cartel, benefit from the higher price umbrella created by the agreement, and slightly undercut the cartel price. This interpretation is also consistent with evidence from gasoline retail markets in Clark and Houde (2014) and Byrne and De Roos (2019), where smaller stations are often the firms that price below the collusive level. The cutoff structure of the model captures this pattern: larger retailers join the cartel, while smaller retailers are more likely to remain outside. The firm-level evidence in Table 5 is consistent with this mechanism. For small retailers, the estimated effects on prices and retail markups are negative but not statistically significant, suggesting that these firms price close to the cartel level while still charging slightly lower prices than larger cartel participants.

Wholesale pricing. Wholesalers anticipate the retail cartel's pricing rule when choosing quantities. This is the key difference from the non-cartel benchmark. When retailers are competitive, the wholesale price is passed through directly to consumers. When retailers form a cartel, the wholesale price affects final demand only through the cartel's optimal retail-price response.

There are two possible regions. If the retail-cartel incentive constraint binds, then

$$p_r^{RC}(p_w) = \bar{p}(K_\Gamma) = D^{-1}(K - \theta K_\Gamma).$$

In this case, the retail price is fixed by the sustainability constraint and final demand is

$$Q^{RC} = D(\bar{p}(K_\Gamma)) = K - \theta K_\Gamma.$$

Since the retail price no longer responds to upstream quantities, the upstream problem does not admit an interior Cournot solution. I therefore use the refinement that, in this boundary region, wholesalers set the wholesale price equal to the retail price,

$$\bar{p}_w^{RC} = \bar{p}(K_\Gamma).$$

The retail cartel then earns zero profit:

$$\Pi_{\Gamma}^{RC} = (\bar{p}(K_{\Gamma}) - \bar{p}_w^{RC}) [D(\bar{p}(K_{\Gamma})) - (K - K_{\Gamma})] = 0.$$

Thus, whenever there exists an interior cartel configuration with strictly positive profit, the retail cartel does not choose a binding configuration. The relevant positive-profit retail-cartel outcomes therefore lie in the interior region.

In the interior region, the Bos–Harrington constraint is slack and the retail cartel charges $p^*(p_w, K_{\Gamma})$. Let $Q = D(p_r^{RC})$ denote final demand and let $P(Q) = D^{-1}(Q)$ denote inverse demand. Rewriting the retail cartel's first-order condition gives the derived inverse demand faced by wholesalers:

$$P_w^{RC}(Q, K_{\Gamma}) = P(Q) + [Q - (K - K_{\Gamma})] P'(Q). \quad (10)$$

This object differs from final inverse demand because the downstream cartel prices over residual demand rather than total market demand.

Wholesalers compete à la Cournot over this derived demand. In a symmetric Cournot equilibrium, total quantity Q^{RC} solves

$$P_w^{RC}(Q^{RC}, K_{\Gamma}) - c + \frac{Q^{RC}}{N_w} \frac{\partial P_w^{RC}(Q^{RC}, K_{\Gamma})}{\partial Q} = 0. \quad (11)$$

The equilibrium wholesale price is therefore

$$p_w^{RC} = c - \frac{Q^{RC}}{N_w} \frac{\partial P_w^{RC}(Q^{RC}, K_{\Gamma})}{\partial Q}, \quad (12)$$

and the corresponding retail price is

$$p_r^{RC} = p_r^{RC}(p_w^{RC}).$$

The retail-cartel case therefore features two margins: wholesalers charge a Cournot markup over marginal cost, and the retail cartel charges a markup over the whole-sale price. However, the final retail price need not exceed the vertically integrated monopoly price, because the competitive fringe reduces the residual demand internalized by the cartel.

Theorem 1. Let $p^M(c)$ denote the vertically integrated monopoly price,

$$p^M(c) \in \arg \max_p (p - c)D(p).$$

Consider a non-fully-inclusive retail cartel, so that $K - K_\Gamma > 0$, and suppose the residual-demand profit function is single-peaked. If

$$K - K_\Gamma \geq (p_w^{RC} - c) [-D'(p^M(c))],$$

then

$$p_r^{RC} \leq p^M(c).$$

Proof. Define

$$H(p; p_w^{RC}, K_\Gamma) = D(p) - (K - K_\Gamma) + (p - p_w^{RC})D'(p).$$

The unconstrained residual-monopoly price satisfies

$$H(p^*; p_w^{RC}, K_\Gamma) = 0.$$

Evaluating H at $p^M(c)$ and using the integrated monopoly first-order condition,

$$D(p^M) + (p^M - c)D'(p^M) = 0,$$

gives

$$H(p^M; p_w^{RC}, K_\Gamma) = (p_w^{RC} - c) [-D'(p^M)] - (K - K_\Gamma).$$

By assumption, this expression is weakly negative. Since the residual-demand profit function is single-peaked, the unconstrained residual-monopoly price satisfies

$$p^*(p_w^{RC}, K_\Gamma) \leq p^M(c).$$

Finally,

$$p_r^{RC} = \min\{p^*(p_w^{RC}, K_\Gamma), \bar{p}(K_\Gamma)\},$$

so

$$p_r^{RC} \leq p^M(c).$$

□

Theorem 1 shows that a retail cartel does not mechanically imply a retail price above the integrated monopoly price. The condition compares two forces. The fringe capacity $K - K_\Gamma$ pushes the retail price downward by reducing the residual demand faced by the cartel. The upstream markup,

$$(p_w^{RC} - c) [-D'(p^M(c))],$$

pushes the retail price upward through double marginalization. When fringe capacity is sufficiently large relative to this upstream distortion, the residual-demand effect dominates and the retail-cartel price remains weakly below $p^M(c)$.

5.5 Double Horizontal Cartel

I now consider the case in which both wholesalers and retailers form cartels, but the two cartels do not coordinate vertically. Relative to the retail-cartel case, the upstream price is now chosen by a wholesale cartel rather than by Cournot competitors. However, the two cartels still choose prices independently. The market therefore features two separate markups: the wholesale cartel charges a markup over marginal cost, and the retail cartel charges a markup over the wholesale price.

As before, I solve the pricing game backwards. For any wholesale price p_w , the downstream cartel chooses the retail price according to the Bos–Harrington pricing rule:

$$p_r^{DH}(p_w) = \min\{p^*(p_w, K_\Gamma), \bar{p}(K_\Gamma)\},$$

where $p^*(p_w, K_\Gamma)$ is the unconstrained residual-monopoly price and

$$\bar{p}(K_\Gamma) = D^{-1}(K - \theta K_\Gamma)$$

is the highest price satisfying the downstream incentive constraint.

If the downstream incentive constraint binds, then

$$p_r^{DH}(p_w) = \bar{p}(K_\Gamma),$$

and final demand is

$$D(\bar{p}(K_\Gamma)) = K - \theta K_\Gamma.$$

The wholesale cartel then solves

$$\max_{p_w \in [c, \bar{p}(K_\Gamma)]} (p_w - c)(K - \theta K_\Gamma).$$

Since the objective is increasing in p_w , the wholesale cartel chooses

$$\bar{p}_w^{DH} = \bar{p}(K_\Gamma).$$

Thus, the wholesale price coincides with the retail price, and the retail cartel earns zero profit:

$$\Pi_\Gamma^{DH} = (\bar{p}(K_\Gamma) - \bar{p}_w^{DH}) [D(\bar{p}(K_\Gamma)) - (K - K_\Gamma)] = 0.$$

As in the retail-cartel case, such a configuration is never selected when a positive-profit interior cartel is feasible. Therefore, any active and profitable double-horizontal cartel lies in the interior region, where the downstream incentive constraint is slack.

In the interior region,

$$p_r^{DH}(p_w) = p^*(p_w, K_\Gamma).$$

The wholesale cartel chooses the wholesale price anticipating this downstream pricing response:

$$p_w^{DH} \in \arg \max_{p_w} (p_w - c) D(p^*(p_w, K_\Gamma)). \quad (13)$$

This problem differs from the wholesale-cartel-only case. When retailers are competitive, the retail price equals the wholesale price, so the wholesale cartel faces demand $D(p_w)$. With a downstream cartel, the wholesale cartel instead faces derived demand through the retail cartel's pricing response, $D(p_r^{DH}(p_w))$.

Whenever the downstream cartel earns positive profits, it charges a positive markup over the wholesale price:

$$p_r^{DH}(p_w) > p_w.$$

Because demand is decreasing,

$$D(p_r^{DH}(p_w)) < D(p_w).$$

Therefore, for any wholesale price $p_w > c$,

$$(p_w - c) D(p_r^{DH}(p_w)) < (p_w - c) D(p_w).$$

Thus, the downstream cartel extracts part of the vertical surplus through its retail markup, but it also reduces final demand and thereby lowers the demand faced by the wholesale cartel relative to the wholesale-cartel-only benchmark.

The comparison with the integrated monopoly price follows the same logic as in Theorem 1. The only difference is that the relevant wholesale price is now p_w^{DH} rather than p_w^{RC} . If

$$K - K_\Gamma \geq (p_w^{DH} - c) [-D'(p^M(c))],$$

then

$$p_r^{DH} \leq p^M(c).$$

If the inequality is reversed, the unconstrained double-horizontal price may exceed the integrated monopoly price, although the actual price may still be capped by the Bos–Harrington constraint:

$$p_r^{DH} = \min\{p^*(p_w^{DH}, K_\Gamma), \bar{p}(K_\Gamma)\}.$$

Hence, double-horizontal collusion does not mechanically imply a final price above the integrated monopoly price. The upstream cartel markup pushes the retail price upward, while the competitive fringe pushes it downward by reducing the residual demand faced by the retail cartel.

Although the final retail price under double-horizontal collusion may be above or below the integrated monopoly price, the double-horizontal structure is inefficient from the perspective of joint industry profits whenever it does not implement $p^M(c)$. The wholesale price affects the division of surplus between the two tiers, but total industry profit depends only on the final retail price and the true marginal cost c .

Theorem 2. Suppose that $D(p)$ is decreasing and that $(p - c)D(p)$ is strictly concave in p . Let

$$p^M(c) \in \arg \max_p (p - c)D(p)$$

denote the integrated monopoly price, and let

$$\Pi^M = (p^M(c) - c)D(p^M(c))$$

denote integrated monopoly profit. Let Π_w^{DH} denote aggregate wholesale profit and let Π_R^{DH} denote aggregate downstream retail profit under the double-horizontal structure. Then

$$\Pi_w^{DH} + \Pi_R^{DH} \leq \Pi^M.$$

Moreover, if $p_r^{DH} \neq p^M(c)$, the inequality is strict. Therefore, there exists a division of integrated monopoly profit that makes the upstream and downstream tiers weakly better off relative to the double-horizontal outcome.

Proof. Under the double-horizontal structure, aggregate wholesale profit is

$$\Pi_w^{DH} = (p_w^{DH} - c)D(p_r^{DH}),$$

and aggregate downstream retail profit is

$$\Pi_R^{DH} = (p_r^{DH} - p_w^{DH})D(p_r^{DH}).$$

Adding these terms gives

$$\Pi_w^{DH} + \Pi_R^{DH} = (p_r^{DH} - c)D(p_r^{DH}).$$

Since $p^M(c)$ maximizes $(p - c)D(p)$,

$$\Pi^M = (p^M(c) - c)D(p^M(c)) \geq (p_r^{DH} - c)D(p_r^{DH}) = \Pi_w^{DH} + \Pi_R^{DH}.$$

Strict concavity implies that the inequality is strict whenever $p_r^{DH} \neq p^M(c)$.

Now let $\alpha \in [0, 1]$ denote the share of integrated monopoly profit assigned to the downstream tier. The downstream tier receives $\alpha\Pi^M$, while the upstream tier receives $(1 - \alpha)\Pi^M$. Both tiers are weakly better off if

$$\alpha\Pi^M \geq \Pi_R^{DH}$$

and

$$(1 - \alpha)\Pi^M \geq \Pi_w^{DH}.$$

Equivalently,

$$\frac{\Pi_R^{DH}}{\Pi^M} \leq \alpha \leq 1 - \frac{\Pi_w^{DH}}{\Pi^M}.$$

This interval is nonempty because

$$\Pi_R^{DH} + \Pi_w^{DH} \leq \Pi^M.$$

Therefore, there exists a division of integrated monopoly profit that makes both tiers weakly better off. If $p_r^{DH} \neq p^M(c)$, the interval has positive length and the division can be chosen so that at least one tier is strictly better off. $\square$

Theorem 2 establishes that, whenever the double-horizontal structure fails to implement the integrated monopoly price, vertical coordination can create additional surplus. The next subsection studies whether this surplus can be realized through a vertical cartel and how it is divided between wholesalers and retailers.

5.6 Vertical Cartel

I now consider the case in which wholesalers and retailers coordinate vertically. In contrast to the double-horizontal cartel, the upstream and downstream cartels no longer choose prices independently. Instead, they jointly determine the retail price and wholesale terms in order to maximize total cartel surplus and then divide this surplus among participants.

The institutional evidence suggests two mechanisms through which vertical coordination may operate. The first is the joint determination of wholesale and retail prices. In vertical cartel cases, wholesalers were alleged to participate in meetings organized by gas-station associations, where retail prices and profit margins were discussed. For example, in Case 1 (08700.003447/2015-40), CADE states that fuel wholesalers participated in Sindipetr leo (gas station union) meetings to discuss

prices and profit margins. This evidence motivates the assumption that, under vertical collusion, wholesale and retail decisions are coordinated rather than chosen independently.

The second mechanism is discrimination in wholesale prices between cartel participants and non-participants. In Case 5 (08700.010769/2014-64), CADE reports that a fuel distributor allegedly differentiated wholesale prices in order to favor retailers participating in the collusive agreement and to induce other gas stations to join the cartel. Motivated by this evidence, I allow wholesalers to charge different wholesale prices to cartel insiders and outsiders.

Vertical discrimination. Under vertical collusion, wholesalers post two wholesale prices: p_w^I for retailers that participate in the cartel and p_w^O for retailers that remain outside the cartel. I assume

$$p_w^O > p_w^I \geq c,$$

so cartel insiders receive preferential wholesale terms, while outsiders face a higher input price.

For an outsider to remain active, it must still be able to undercut the cartel price by an arbitrarily small amount $\xi > 0$. Therefore, the outsider wholesale price must satisfy

$$p_w^O \leq p_r^V - \xi. \tag{14}$$

Taking the limit as $\xi \rightarrow 0$, the highest feasible outsider wholesale price is $p_w^O = p_r^V$. Relative to the horizontal retail cartel, vertical discrimination makes non-participation less attractive: an outsider not only remains outside the profit-sharing agreement but also faces a higher wholesale price.

Proposition 1. Under vertical collusion, all retailers participate in the cartel.

Proof. Consider any retailer i that remains outside the cartel. By (14), the highest feasible outsider wholesale price satisfies $p_w^O = p_r^V$ as $\xi \rightarrow 0$. Therefore, an outsider that sells at the cartel price earns zero per-unit margin:

$$p_r^V - p_w^O = 0.$$

By contrast, a retailer that joins the cartel pays $p_w^I < p_r^V$, and therefore earns a strictly positive per-unit margin:

$$p_r^V - p_w^I > 0.$$

Given a strictly positive allocation of cartel surplus, joining strictly dominates remaining outside. Since this argument applies to every retailer, all retailers participate in the cartel under vertical collusion. $\square$

Proposition 1 shows that vertical discrimination eliminates the competitive fringe. In the horizontal retail-cartel case, the cartel is generally not fully inclusive because small retailers may prefer to remain outside and sell at the cartel price. Under vertical collusion, this outside option is weakened by the higher outsider wholesale price. As a result, the vertical cartel induces full downstream participation.

Given full participation, the vertical cartel internalizes the entire vertical margin between the final retail price and true marginal cost c . Therefore, the retail price is chosen to maximize total industry profit:

$$p^M(c) \in \arg \max_p (p - c)D(p). \quad (15)$$

Thus, under vertical collusion,

$$p_r^V = p^M(c).$$

The insider wholesale price p_w^I does not determine total industry profit. Instead, it determines how monopoly surplus is divided between the wholesale cartel and the retail cartel. At the monopoly price $p^M(c)$, total quantity is $D(p^M(c))$. The aggregate profit of the wholesale cartel is

$$\Pi_w^V(p_w^I) = D(p^M(c))(p_w^I - c),$$

while the aggregate profit of the retail cartel is

$$\Pi_r^V(p_w^I) = D(p^M(c))(p^M(c) - p_w^I).$$

I model the determination of p_w^I as a Nash bargaining problem between the upstream and downstream cartels. The disagreement point is the best feasible non-vertical outcome available to wholesalers, conditional on retailers having formed a downstream cartel. This conditioning is important. If the retail cartel is profitable, retailers prefer forming the cartel to reverting to Bertrand competition, which yields zero retail profits. Therefore, the relevant disagreement outcome is not whether retailers collude, but whether wholesalers also collude independently. In the absence of vertical coordination, wholesalers can either remain non-cooperative, generating the retail-horizontal outcome, or form an upstream cartel that operates independently of the retail cartel, generating the double-horizontal outcome. Let

$$\mathcal{O} = \{RH, DH\}$$

denote the set of feasible no-vertical cartel structures, where RH is the retail-horizontal cartel with non-cooperative Cournot wholesalers and DH is the double-horizontal cartel. The double-horizontal outcome belongs to $\mathcal{O}$ only if the upstream cartel is both sustainable and individually rational for wholesalers. The disagreement regime is

$$o^* \in \arg \max_{o \in \mathcal{O}} \Pi_w^o. \quad (16)$$

The corresponding disagreement payoff vector is

$$(\Pi_r^0, \Pi_w^0) = (\Pi_r^{o^*}, \Pi_w^{o^*}). \quad (17)$$

Thus, the retailer and wholesaler disagreement payoffs are always taken from the same fallback regime. If the selected fallback is the retail-horizontal outcome, then $(\Pi_r^0, \Pi_w^0) = (\Pi_r^{RH}, \Pi_w^{RH})$. If the selected fallback is the double-horizontal outcome, then $(\Pi_r^0, \Pi_w^0) = (\Pi_r^{DH}, \Pi_w^{DH})$.

The Nash product is therefore written in terms of gains relative to the selected no-vertical disagreement regime:

$$W(p_w^I) = [D(p^M(c))(p_w^I - c) - \Pi_w^0]^{1-\lambda} [D(p^M(c))(p^M(c) - p_w^I) - \Pi_r^0]^\lambda, \quad (18)$$

where $\lambda \in (0, 1)$ is the bargaining weight of the retail cartel. The insider wholesale price p_w^I is not chosen to determine the final retail price, which is already set at $p^M(c)$. Instead, it allocates monopoly surplus between the two sides of the market.

Theorem 2 implies that the feasible set of Pareto-improving splits is nonempty whenever the selected no-vertical disagreement regime does not implement the integrated monopoly price. The disagreement payoffs (Π_r^0, Π_w^0) determine each side's bargaining leverage. A higher wholesaler payoff in the disagreement regime strengthens the upstream tier's outside option and requires a larger share of monopoly surplus to be allocated to wholesalers. Similarly, a higher retail payoff in the disagreement regime strengthens the downstream tier's outside option and increases the share of surplus that must be allocated to retailers. The bargaining problem therefore divides only the incremental surplus generated by vertical coordination relative to the selected no-vertical fallback.

The vertical cartel must also satisfy incentive constraints. First, the upstream cartel must satisfy the wholesale-cartel incentive constraint,

$$\delta \geq 1 - \frac{1}{N_w},$$

so that no individual wholesaler wants to deviate from the upstream agreement. Second, the downstream cartel incentive constraint (7), evaluated at $p^M(c)$ and full

participation $K_\Gamma = K$, requires

$$D(p^M(c)) \geq (1 - \theta)K,$$

or equivalently,

$$\theta \geq 1 - \frac{D(p^M(c))}{K}. \quad (19)$$

This condition requires retailers to be sufficiently patient for the full downstream cartel to be sustainable at the monopoly price.

Third, the retail cartel must not deviate from the vertical agreement after wholesalers charge the insider wholesale price p_w^I . Let $p_r^D(p_w^I)$ denote the retail cartel's best one-period deviation price:

$$p_r^D(p_w^I) \in \arg \max_{p_r} D(p_r)(p_r - p_w^I),$$

and let

$$\Pi_r^D(p_w^I) = D(p_r^D(p_w^I))(p_r^D(p_w^I) - p_w^I).$$

The downstream incentive constraint is

$$\frac{1}{1 - \theta} \Pi_r^V \geq \Pi_r^D(p_w^I) + \frac{\theta}{1 - \theta} \Pi_r^0.$$

Equivalently,

$$\theta \geq \frac{\Pi_r^D(p_w^I) - \Pi_r^V}{\Pi_r^D(p_w^I) - \Pi_r^0}. \quad (20)$$

Fourth, wholesalers must not deviate from the negotiated wholesale terms. Given that the retail cartel charges $p^M(c)$, the optimal one-period deviation for a wholesaler is to set the wholesale price equal to $p^M(c)$ and capture the entire vertical margin. Hence, the one-period deviation payoff is Π^M . After a deviation, the market reverts to the selected no-vertical disagreement regime o^* , yielding continuation payoff Π_w^0 . The upstream incentive constraint is therefore

$$\frac{1}{1 - \delta} \Pi_w^V \geq \Pi^M + \frac{\delta}{1 - \delta} \Pi_w^0.$$

Equivalently,

$$\delta \geq \frac{\Pi^M - \Pi_w^V}{\Pi^M - \Pi_w^0}. \quad (21)$$

Putting these conditions together, the insider wholesale price solves

$$\begin{aligned} \max_{p_w^I} & [D(p^M(c))(p_w^I - c) - \Pi_w^0]^{1-\lambda} [D(p^M(c))(p^M(c) - p_w^I) - \Pi_r^0]^\lambda \\ \text{s.t.} \quad & \theta \geq \max \left\{ 1 - \frac{D(p^M(c))}{K}, \frac{\Pi_r^D(p_w^I) - \Pi_r^V}{\Pi_r^D(p_w^I) - \Pi_r^0} \right\}, \\ & \delta \geq \max \left\{ 1 - \frac{1}{N_w}, \frac{\Pi^M - \Pi_w^V}{\Pi^M - \Pi_w^0} \right\}. \end{aligned} \quad (22)$$

Assuming that none of the incentive constraints binds, the Nash bargaining solution gives the insider wholesale price

$$p_w^{I,V} = c + \frac{\Pi_w^0 + (1 - \lambda) [\Pi^M - \Pi_w^0 - \Pi_r^0]}{D(p^M(c))}. \quad (23)$$

Equivalently, the vertical cartel allocates the incremental surplus from coordination according to the bargaining weights:

$$\Pi_w^V = \Pi_w^0 + (1 - \lambda) [\Pi^M - \Pi_w^0 - \Pi_r^0],$$

and

$$\Pi_r^V = \Pi_r^0 + \lambda [\Pi^M - \Pi_w^0 - \Pi_r^0].$$

The intuition is that the retail price $p^M(c)$ determines the size of total surplus, while the insider wholesale price $p_w^{I,V}$ determines how this surplus is divided. A higher $p_w^{I,V}$ transfers surplus from retailers to wholesalers; a lower $p_w^{I,V}$ transfers surplus from wholesalers to retailers. The vertical cartel does not divide total monopoly profit only according to λ . Instead, it first guarantees each side its selected no-vertical disagreement payoff and then divides only the incremental surplus,

$$\Pi^M - \Pi_w^0 - \Pi_r^0,$$

according to the bargaining weights.

This result is consistent with the reduced-form evidence at both the city level, reported in Table 4, and the station level, reported in Table 5. In both sets of estimates, vertical cartels generate larger retail-price effects than horizontal cartels. Retail markups also increase more under vertical cartels, indicating that upstream participation allows the cartel to sustain higher downstream margins. The model rationalizes this pattern by formalizing the mechanisms documented in the legal

evidence: wholesalers participate in cartel pricing decisions and can discriminate between cartel insiders and outsiders through wholesale terms. By weakening the outside option of non-participating retailers, vertical coordination relaxes the constraint imposed by the competitive fringe and allows the cartel to sustain higher retail prices.

The evidence on wholesale prices is more muted. The estimates point to a positive increase in wholesale prices under vertical cartels, but the effect is less precisely estimated. This is not inconsistent with the model. The model does not imply that the insider wholesale price must necessarily increase. The reason is that the wholesale price serves two roles: it affects the retail price indirectly, but it also allocates surplus between wholesalers and retailers. If the disagreement outcome involves double marginalization, either because wholesalers compete imperfectly or because upstream and downstream cartels operate independently, vertical coordination can increase joint surplus even with an insider wholesale price close to, or below, the non-cartel benchmark. Thus, the level of the wholesale price depends on disagreement payoffs and bargaining leverage. When wholesalers have relatively low bargaining power, the insider wholesale price may increase only slightly, or even decrease, while vertical coordination still raises total cartel surplus.

Finally, the station-level results in Table 5 are consistent with the model’s price-discrimination mechanism. In vertical-cartel cases, small stations charge retail prices similar to those of larger stations, suggesting that the cartel achieves relatively homogeneous retail pricing. However, small stations experience smaller retail-markup gains because they face higher wholesale prices. This pattern is consistent with wholesalers using wholesale terms to discipline retailers with stronger incentives to deviate. Since smaller stations are more likely to benefit from undercutting the cartel price or remaining outside the agreement, they are also the stations for which discriminatory wholesale pricing is most relevant. In this sense, the firm-level evidence supports the model’s central mechanism: vertical participation helps sustain downstream coordination by reducing the profitability of deviation through wholesale-price discipline.

5.7 Cartel Formation

Having characterized the pricing outcome under each market structure, I now summarize the conditions under which each structure is implemented. I solve the formation stage backwards. Retailers choose whether to form a downstream cartel, anticipating the wholesale pricing regime and the possibility of vertical coordination. Wholesalers then decide whether to form an upstream cartel and, when both tiers

collude, whether to coordinate vertically.

Let RH denote the retail-horizontal outcome, WC the wholesale-cartel outcome, DH the double-horizontal outcome, and V the vertical-cartel outcome. The wholesale cartel can be sustained only if the incentive constraint in (6) holds:

$$\delta \geq 1 - \frac{1}{N_w}. \quad (24)$$

This condition is necessary for both the wholesale-cartel and double-horizontal outcomes, and it also enters the feasibility of vertical collusion.

For each wholesale regime $o \in \{RH, DH\}$, let

$$\Pi_r^{o,*} = \max_{\Gamma \in \mathcal{F}^o} \Pi_r^o(\Gamma)$$

denote the maximum downstream cartel profit over feasible retail-cartel configurations, and let $\Gamma^{o,*}$ be the maximizing cartel. Define the corresponding wholesaler profit as

$$\Pi_w^{o,*} = \Pi_w^o(\Gamma^{o,*}).$$

Thus, $\Pi_r^{RH,*}$ and $\Pi_w^{RH,*}$ are the retailer and wholesaler profits when retailers collude and wholesalers remain non-cooperative, while $\Pi_r^{DH,*}$ and $\Pi_w^{DH,*}$ are the corresponding profits when both tiers form horizontal cartels without vertical coordination.

The no-vertical disagreement regime for vertical bargaining is the feasible no-vertical outcome preferred by wholesalers:

$$o^* \in \arg \max_{o \in \mathcal{O}} \Pi_w^o, \quad \mathcal{O} \subseteq \{RH, DH\}.$$

The outcome RH belongs to $\mathcal{O}$ whenever $\Pi_r^{RH,*} > 0$. The outcome DH belongs to $\mathcal{O}$ only if upstream collusion is sustainable and yields higher wholesaler profits than the retail-horizontal outcome:

$$\delta \geq 1 - \frac{1}{N_w} \quad \text{and} \quad \Pi_w^{DH,*} \geq \Pi_w^{RH,*}.$$

The disagreement payoffs are

$$(\Pi_r^0, \Pi_w^0) = (\Pi_r^{o^*}, \Pi_w^{o^*}).$$

A vertical agreement with insider wholesale price p_w^I is feasible if it solves the Nash bargaining problem and satisfies

$$\theta \geq \max \left\{ 1 - \frac{D(p^M(c))}{K}, \frac{\Pi_r^D(p_w^I) - \Pi_r^V}{\Pi_r^D(p_w^I) - \Pi_r^0} \right\}, \quad (25)$$

and

$$\delta \geq \max \left\{ 1 - \frac{1}{N_w}, \frac{\Pi^M - \Pi_w^V}{\Pi^M - \Pi_w^0} \right\}. \quad (26)$$

The first condition requires full downstream participation and no profitable downstream deviation. The second requires upstream cartel sustainability and no profitable upstream deviation.

The implemented cartel structure is characterized as follows.

1. **Non-cartel benchmark.** If no feasible downstream cartel generates positive profit,

$$\Pi_r^{RH,*} \leq 0,$$

and upstream collusion cannot be sustained,

$$\delta < 1 - \frac{1}{N_w},$$

then neither tier forms a cartel.

2. **Wholesale cartel.** If no feasible downstream cartel generates positive profit,

$$\Pi_r^{RH,*} \leq 0,$$

and upstream collusion can be sustained,

$$\delta \geq 1 - \frac{1}{N_w},$$

then wholesalers form an upstream cartel while retailers remain competitive.

3. **Retail-horizontal cartel.** If a feasible downstream cartel generates positive profit,

$$\Pi_r^{RH,*} > 0,$$

and vertical coordination is not feasible, the retail-horizontal cartel is implemented when either upstream collusion is not sustainable,

$$\delta < 1 - \frac{1}{N_w},$$

or wholesalers prefer the profits generated under the retail-horizontal outcome to those generated by upstream collusion:

$$\Pi_w^{RH,*} \geq \Pi_w^{WC}$$

and, when the double-horizontal outcome is feasible,

$$\Pi_w^{RH,*} \geq \Pi_w^{DH,*}.$$

In this case, retailers form $\Gamma^{RH,*}$, while wholesalers remain non-cooperative.

4. **Double-horizontal cartel.** If upstream collusion is sustainable,

$$\delta \geq 1 - \frac{1}{N_w},$$

a profitable downstream cartel exists under upstream collusion,

$$\Pi_r^{DH,*} > 0,$$

vertical coordination is not feasible, and wholesalers prefer double-horizontal collusion to the retail-horizontal outcome,

$$\Pi_w^{DH,*} > \Pi_w^{RH,*},$$

then both tiers form horizontal cartels but do not coordinate vertically.

5. **Vertical cartel.** If a downstream cartel can be formed, upstream collusion is sustainable, and there exists an insider wholesale price p_w^I satisfying (25) and (26), then the two tiers coordinate vertically. The retail price is

$$p_r^V = p^M(c),$$

the insider wholesale price is determined by Nash bargaining, and Proposition 1 implies full downstream participation.

This characterization highlights that upstream collusion requires both sustainability and profitability. A retail-horizontal cartel can arise either because upstream collusion is not sustainable or because wholesalers earn more when retailers collude downstream and wholesalers remain non-cooperative. Conditional on both tiers being able to collude, vertical coordination is implemented only when the incremental surplus relative to the selected no-vertical disagreement regime can be divided in a way that satisfies both tiers' no-deviation constraints.

5.8 Mechanisms and Empirical Predictions

The model shows that the mechanisms documented in the legal records are sufficient to rationalize the main empirical patterns in the reduced-form evidence. In the horizontal-cartel case, downstream collusion is constrained by the participation incentives of retailers and by the presence of a competitive fringe. Retailers outside the agreement can undercut the cartel price and limit the residual demand served by cartel members. As a result, a purely horizontal retail cartel need not be fully inclusive and may sustain only a limited increase in retail prices and retail markups.

Vertical cartels operate through a different mechanism. When wholesalers participate in the agreement, they do not simply add an upstream cartel to a downstream cartel. Instead, wholesalers and retailers jointly determine prices and divide the resulting surplus. This is captured in the model through Nash bargaining between the two tiers. Moreover, wholesalers can use their vertical relationship with retailers to discriminate between cartel insiders and outsiders. By offering more favorable wholesale terms to participating retailers and less favorable terms to non-participants, wholesalers reduce the profitability of remaining outside the cartel or deviating from the agreement. This weakens the competitive fringe, supports broader downstream participation, and allows the cartel to sustain higher retail prices.

These mechanisms recover the ranking found in the reduced-form results. Both the city-level and station-level estimates show that vertical cartels generate larger increases in retail prices and retail markups than horizontal cartels. The model rationalizes this result because vertical coordination allows the cartel to internalize the full vertical margin and to relax the downstream participation constraint that limits horizontal collusion. At the same time, the model does not require a large or precisely estimated increase in the insider wholesale price. Under vertical coordination, the wholesale price serves primarily to allocate surplus between wholesalers and retailers, while the final retail price is determined by the joint maximization of vertical surplus. Therefore, the muted evidence on wholesale prices is consistent with the model: vertical participation can substantially increase retail prices and retail markups even when the observed wholesale-price response is small.

Overall, the model provides a bridge between the documentary evidence and the empirical estimates. The legal records identify the institutional mechanisms through which wholesalers facilitated downstream coordination: participation in price-setting discussions and discriminatory wholesale terms. Once these mechanisms are introduced into a capacity-constrained model of cartel formation, the model reproduces the central reduced-form finding that vertical cartels have larger effects than horizontal cartels. This supports the interpretation that the higher overcharges observed

in vertical cases are not merely the result of stronger horizontal coordination, but reflect the additional ability of wholesalers to discipline retailers and sustain more profitable downstream collusion.

6 Model Calibration and Counterfactual Mechanisms

I complement the reduced-form evidence with a calibrated numerical exercise based on the theoretical model. The goal is not to structurally estimate cartel conduct or to quantify damages. Instead, the exercise asks whether the model can reproduce the qualitative ranking documented in the data under a transparent parameterization disciplined by observable market characteristics.

The reduced-form results show that vertical cartels generate larger effects than horizontal cartels on retail prices and retail markups, with weaker but consistent evidence of larger effects on wholesale markups. The calibration reproduces this ranking. Vertical coordination generates larger retail-price, retail-markup, and wholesale-markup effects than a purely horizontal downstream cartel. The exercise therefore provides a mechanism-based interpretation of the empirical evidence: horizontal retail cartels are constrained by fringe capacity and by the participation incentives of marginal retailers, while vertical cartels can weaken the outside option of non-participating retailers and divide the additional surplus through wholesale prices.

The calibration should be interpreted as a stylized mechanism exercise. Two data limitations prevent a full structural estimation. First, station-level quantities are not observed, which prevents the estimation of case-specific demand elasticities and station-level market shares. Second, the true composition of each cartel is unobserved. Legal records identify investigated or convicted firms, but these sets need not coincide with the set of firms that actually participated in the agreement. The model also abstracts from spatial differentiation, heterogeneous local demand, and heterogeneity in wholesaler market shares. For these reasons, I focus on the ranking of outcomes across cartel structures, rather than on matching the magnitude of the reduced-form estimates.

6.1 Calibration Strategy

I normalize the non-cartel benchmark retail price and total quantity to one:

$$p_0 = 1, \quad Q_0 = 1.$$

Demand is linear,

$$D(p) = A - Bp,$$

with $B > 0$. The parameters A and B are pinned down by the benchmark quantity and the benchmark market-demand elasticity. At p_0 ,

$$\varepsilon_0 = \frac{D'(p_0)p_0}{D(p_0)} = -\frac{Bp_0}{Q_0}.$$

Thus,

$$B = -\varepsilon_0 \frac{Q_0}{p_0}, \quad A = Q_0 + Bp_0.$$

Given $p_0 = Q_0 = 1$, this becomes

$$B = -\varepsilon_0, \quad A = 1 - \varepsilon_0.$$

In the baseline parameterization, I set $\varepsilon_0 = -0.399$, following Santos (2013). This elasticity is interpreted as the aggregate market-demand elasticity, not as the residual elasticity faced by an individual station.

The baseline exercise uses the Brasília/DF cartel case, Case 8, to discipline market structure. Retail capacity is measured using the best available station-level gasoline storage-capacity information. I preserve the empirical distribution of capacity across stations and rescale aggregate capacity to

$$K = 1.10Q_0.$$

This introduces moderate excess capacity, allowing fringe capacity to discipline downstream collusion without making cartel formation infeasible by construction. For the upstream market, I set $N_w = 3$, consistent with evidence in Chaves and Duarte (2025) that three wholesalers account for the main relevant upstream activity in this market.

Marginal cost is recovered from the non-cartel benchmark. In the benchmark, retailers are competitive and wholesalers compete à la Cournot, so

$$p_r^N = p_w^N = p_0.$$

Given N_w , p_0 , Q_0 , and the demand elasticity, marginal cost is pinned down by the symmetric Cournot first-order condition:

$$c = p_0 + \frac{Q_0}{N_w} P'(Q_0),$$

where $P(\cdot) = D^{-1}(\cdot)$. Since $P'(Q_0) < 0$, the recovered marginal cost is below the normalized benchmark price. The integrated monopoly price is then

$$p^M = \frac{1}{2} \left(\frac{A}{B} + c \right).$$

Retail cartels are formed using an m -cutoff rule. Retailers are ordered by capacity from largest to smallest, and a candidate cartel of size m contains the m largest retailers. I solve the model for all candidate cutoff cartels,

$$m = 1, \dots, N_r - 1.$$

For each candidate cartel, I impose the participation and stability conditions from the model: the marginal insider must prefer to remain inside the cartel, the marginal outsider must prefer to remain outside, and the selected outcome must lie in the interior region where the Bos–Harrington constraint does not bind. Among feasible cartels, the selected cartel maximizes total insider retail profit.

The baseline sets both discount factors equal to

$$\delta = 0.95, \quad \theta = 0.95.$$

These values represent patient firms and ensure that incentive constraints are not mechanically violated in the baseline case. I compare five conduct regimes: the non-cartel benchmark, a wholesale cartel, a retail-horizontal cartel, a double-horizontal cartel, and a vertical cartel. In the vertical cartel, the retail price is set at the integrated monopoly price,

$$p^V = p^M,$$

and the insider wholesale price allocates monopoly surplus between wholesalers and retailers. Rather than fixing a single bargaining weight, I compute the set of feasible bargaining weights $\lambda \in [0, 1]$ satisfying the participation and incentive constraints. For reported vertical-cartel outcomes, I use the median feasible bargaining weight unless stated otherwise.

Finally, when comparing implemented structures, I first impose feasibility and participation constraints for both tiers. Among feasible structures, I select the outcome preferred by retailers, consistent with the formation timing in which retailers choose the downstream cartel configuration. For the vertical cartel, retailer profit is increasing in the bargaining weight λ , so the retailer-preferred vertical outcome corresponds to the largest feasible λ .

6.2 Baseline Counterfactuals

Table 6 reports the profit implications of the baseline calibration. The wholesale-cartel-only arrangement increases wholesale profit by 33.3 percent relative to the benchmark. Since retailers remain competitive in this case, the increase in wholesale profit is also the increase in total producer profit.

Table 6: Cartel Profit Changes

Statistic	Cartel configuration			
	Wholesale only	Retail only	Double horiz.	Vertical
Retail-cartel capacity share (%)	—	29.9	74.6	100.0
Wholesale profit change (%)	33.3	24.5	-2.5	25.4
Total producer profit change (%)	33.3	24.6	2.1	33.3
Retail profit / benchmark producer profit (%)	0.0	0.1	4.6	7.9

Notes: The table reports model-generated outcomes for Case 8. The first row gives the share of total retail capacity included in the retail cartel. Wholesale profit and total producer profit are reported as percentage changes relative to the competitive benchmark. Total producer profit is the sum of wholesale and retail profits. Retail profit is reported as a percentage of benchmark total producer profit because benchmark retail profit is zero. The dash indicates that no retail cartel is formed.

The table illustrates how upstream and downstream incentives interact. When only retailers coordinate, the retail cartel includes 29.9 percent of total retail capacity. When wholesalers also coordinate but do not coordinate vertically with retailers, the retail cartel expands to 74.6 percent of total capacity. Thus, upstream conduct affects downstream cartel formation by changing the profitability of retail participation. However, the double-horizontal outcome also reveals a conflict between tiers: retail profit increases, but wholesale profit falls relative to the benchmark because the downstream cartel reduces final demand.

Vertical coordination resolves this conflict by allowing the two tiers to maximize joint surplus and divide it through the insider wholesale price. In the baseline calibration, vertical coordination generates higher total producer surplus than the retail-horizontal cartel. Most of this surplus accrues upstream, but retailers also receive enough surplus to satisfy their participation constraints. This pattern is consistent with the theoretical mechanism: vertical coordination can make both tiers willing to cooperate even when separate horizontal collusion creates inefficient double marginalization.

Table 7 reports the corresponding price and markup effects. The calibrated magnitudes are larger than the reduced-form estimates and should not be interpreted as quantitative estimates of cartel overcharges. This reflects the stylized nature of the model, the absence of spatial differentiation, and the assumption that vertical coordination implements the integrated monopoly price. The relevant comparison is

therefore the ranking across cartel structures.

Table 7: Cartel Price and Markup Changes

Statistic	Cartel configuration			
	Wholesale only	Retail only	Double horiz.	Vertical
Retail price change (%)	83.5	52.9	149.6	83.5
Wholesale price change (%)	83.5	48.3	118.5	73.6
Wholesale markup change (%)	100.0	57.8	141.9	88.1
Total markup change (%)	100.0	63.3	179.1	100.0
Retail markup / benchmark total markup (%)	0.0	5.5	37.2	11.9

Notes: The table reports model-generated price and markup changes for Case 8. Retail and wholesale prices are reported as percentage changes relative to the competitive benchmark. Wholesale markup is defined as $p_w - c$, total markup as $p_r - c$, and implied retail markup as $p_r - p_w$. The implied retail markup is reported as a percentage of benchmark total markup because the benchmark retail markup may be zero.

The ranking is consistent with the reduced-form evidence. The retail-horizontal cartel raises retail prices and retail markups, but its effect is limited by incomplete downstream participation. In the baseline, the endogenous retail cartel covers less than one-third of total retail capacity. Vertical coordination eliminates the competitive fringe, induces full downstream participation, and generates larger retail-price, retail-markup, and wholesale-markup effects than the retail-horizontal cartel.

The double-horizontal cartel generates the largest price and markup increases. This outcome should be interpreted as a counterfactual benchmark rather than as the empirical counterpart of the observed vertical cases. Because the upstream and downstream cartels choose prices independently, the market features two separate collusive margins. The resulting double marginalization can generate very high final prices while reducing total industry surplus relative to integrated vertical coordination.

6.3 Welfare and Rent Sharing

Table 8 reports welfare effects. All cartel configurations reduce consumer surplus and total welfare relative to the benchmark. The retail-horizontal cartel generates the smallest welfare loss because partial participation limits the price increase. The wholesale-cartel and vertical-cartel outcomes generate larger consumer losses because both implement the integrated monopoly price in this calibration. The double-horizontal cartel generates the largest welfare loss because separate upstream and downstream collusion creates the largest pricing distortion.

Table 9 reports the feasible bargaining range under vertical coordination. Although the feasible range of λ is wide, the implied allocation of monopoly surplus

Table 8: Cartel Welfare Changes

Statistic	Cartel configuration			
	Wholesale only	Retail only	Double horiz.	Vertical
Consumer surplus	-55.6	-37.7	-83.8	-55.6
Producer surplus	33.3	24.6	2.1	33.3
Total welfare	-20.0	-12.8	-49.4	-20.0

Notes: The table reports model-generated welfare changes for Case 8 under alternative cartel configurations. Consumer surplus, producer surplus, and total welfare are reported as percentage changes relative to the competitive benchmark. Producer surplus is the sum of wholesale and retail profits. Total welfare is the sum of consumer and producer surplus.

is highly asymmetric. Retailers receive between 1.4 and 5.9 percent of monopoly profits, while wholesalers receive most of the surplus.

Table 9: Implied Division of Monopoly Profit

Statistic	Lower bound	Median	Upper bound
λ	0.1963	0.5475	0.8988
Retailer share of monopoly profit (%)	1.4	3.6	5.9
Wholesaler share of monopoly profit (%)	98.6	96.4	94.1

Notes: The table reports the implied division of monopoly profit across retailers and wholesalers for the lower-bound, median, and upper-bound values of λ . Retailer and wholesaler shares sum to 100 percent within each column.

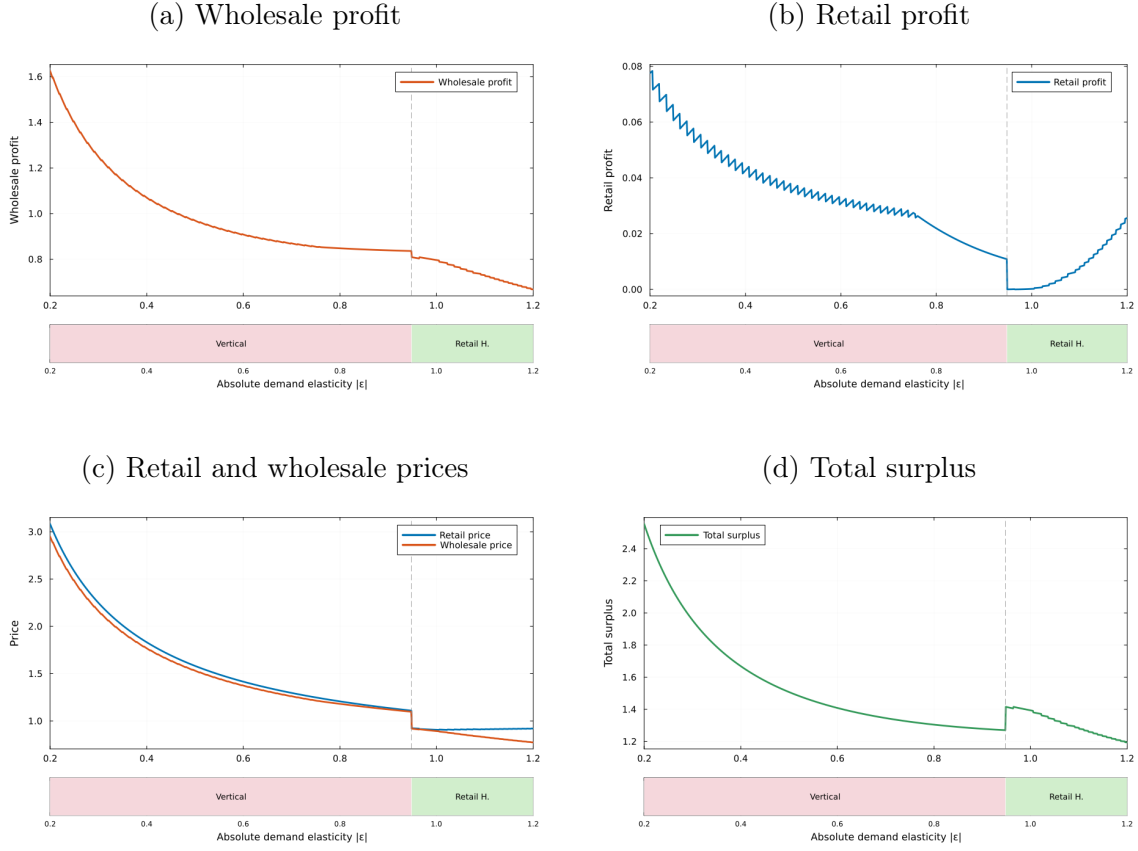
This allocation reflects the bargaining leverage of wholesalers in the model. Retailers need access to wholesale supply to sustain coordinated retail pricing, while wholesalers can affect downstream participation through the wholesale price. The small retailer share does not imply that retailers are worse off from vertical coordination. Instead, the vertical agreement guarantees retailers their selected no-vertical disagreement payoff and then allocates part of the incremental surplus generated by coordination.

6.4 Sensitivity to Demand Elasticity

I next examine how the implemented cartel structure changes with the elasticity of market demand. I keep the baseline market structure fixed, including the empirical distribution of retail capacity, the number of wholesalers, and the recovered marginal cost, and vary only the demand elasticity. For each elasticity value, I recalibrate the linear demand curve to keep the benchmark point fixed at $p_0 = Q_0 = 1$, solve the model for all cartel structures, and select the feasible outcome preferred by retailers subject to wholesaler participation.

Figure 3 shows that demand elasticity affects both the intensity of cartel pricing

Figure 3: Sensitivity of implemented cartel outcomes to demand elasticity



Notes: The figure reports the model-selected outcome as the absolute value of demand elasticity varies. The baseline market structure is held fixed. For each elasticity value, the linear demand curve is recalibrated to keep $p_0 = Q_0 = 1$, while marginal cost remains fixed. The implemented outcome is the feasible cartel structure preferred by retailers, subject to wholesaler participation. The shaded bar reports the selected cartel type: pink denotes vertical coordination and green denotes a retail-horizontal cartel. The dashed vertical line marks the elasticity at which the implemented cartel switches from vertical coordination to a retail-horizontal cartel. The vertical-cartel outcome is computed using the median feasible bargaining parameter.

and the type of cartel that is implemented. For low and intermediate elasticities, the selected outcome is vertical coordination. Demand is sufficiently inelastic that the gains from coordinating the two tiers and charging the integrated monopoly price dominate the no-vertical alternatives. As demand becomes more elastic, monopoly pricing becomes less attractive because the associated quantity loss becomes larger. Around $|\varepsilon| = 0.95$, the implemented cartel switches from vertical coordination to a retail-horizontal cartel.

The price panel illustrates this mechanism. In the vertical-cartel region, retail and wholesale prices decline smoothly as demand becomes more elastic. After the regime switch, the selected retail-horizontal cartel charges a lower retail price than the vertical cartel would, because it internalizes only residual demand and remains disciplined by fringe capacity. The profit panels show the distributional implications: wholesale profit falls sharply when the model switches away from vertical coordination, while retail profit changes less smoothly because the feasible bargaining set depends on discrete downstream deviation cartels. The total-surplus panel shows that the regime switch raises total surplus by moving the market away from integrated monopoly pricing and toward a lower-price retail-horizontal cartel.

Overall, the sensitivity exercise shows that the model can generate both vertical and horizontal cartel outcomes under the same institutional framework. Vertical cartels are more likely to be selected when demand is relatively inelastic and the gains from integrated pricing are large. Retail-horizontal cartels are more likely when demand is sufficiently elastic that monopoly pricing reduces quantity too much. This comparative static reinforces the main mechanism: vertical coordination is most valuable when eliminating the competitive fringe and coordinating wholesale terms generates enough surplus to compensate both tiers.

7 Conclusion

This paper studies how vertical participation changes both the organization and the consequences of collusion in gasoline retail markets. I combine documentary evidence from Brazilian antitrust cases, reduced-form estimates using retail and wholesale price data, and a theoretical model of capacity-constrained downstream collusion with upstream participation. The central argument is that vertical cartels are not simply horizontal retail cartels with an upstream tier added. When wholesalers participate in the collusive arrangement, they can affect both the set of retailers that participate and the prices used to sustain the agreement.

The legal records point to two mechanisms that distinguish vertical cartels from

purely horizontal retail cartels. First, wholesalers participate directly in discussions over retail prices and retail margins. Second, wholesalers can use their vertical relationship with retailers to discipline the agreement by offering more favorable wholesale terms to participating stations and less favorable terms to outsiders or deviating retailers. The model formalizes these mechanisms. In a horizontal retail cartel, collusion is constrained by fringe capacity and by the participation incentives of smaller retailers. Some stations may remain outside the agreement, undercut the cartel price, and limit the residual demand internalized by cartel members. Under vertical collusion, discriminatory wholesale prices weaken this outside option, support broader downstream participation, and allow the cartel to sustain higher retail prices and retail markups. The insider wholesale price then serves primarily to divide the surplus generated by vertical coordination between wholesalers and retailers.

The empirical results are consistent with these predictions. Using eight Brazilian gasoline cartel cases investigated by CADE, I combine weekly retail and wholesale prices with station-level characteristics and information on cartel organization. A two-step matching difference-in-differences design shows that vertical cartels generate larger effects than horizontal cartels on retail prices and retail markups. The evidence on wholesale prices and wholesale markups is more muted, but this is not inconsistent with the model. Under vertical coordination, the wholesale price does not only determine marginal costs downstream; it also allocates surplus between the two tiers. Therefore, vertical cartels can substantially increase retail prices and retail margins even when the observed response of wholesale prices is small or imprecisely estimated.

The station-level results provide additional support for the proposed mechanism. In horizontal cartels, smaller retailers are more likely to act as fringe firms that constrain the cartel price. In vertical-cartel cases, small stations are brought closer to the cartel pricing arrangement, but their retail-markup gains are limited by less favorable wholesale terms. This pattern is consistent with wholesalers using wholesale prices to reduce the profitability of deviation or non-participation among retailers with stronger incentives to remain outside the agreement. Thus, the firm-level heterogeneity helps connect the documentary evidence on wholesale-price discrimination to the aggregate result that vertical cartels generate larger retail-price effects.

The calibrated model complements the reduced-form evidence by showing that the mechanisms documented in the legal records are sufficient to reproduce the qualitative ranking found in the data. The calibration is not a structural estimation and is not intended to match the exact magnitude of the estimated effects. Instead, it asks whether a model disciplined by observable market characteristics can generate larger effects for vertical cartels than for horizontal retail cartels. Once the model incorporates capacity constraints, fringe retailers, wholesale-price discrimina-

tion, and Nash bargaining between wholesalers and retailers, vertical coordination generates larger retail-price and retail-markup effects than horizontal downstream collusion. The exercise therefore provides a mechanism-based interpretation of the reduced-form results.

The findings have broader implications for antitrust enforcement. Vertical arrangements are often evaluated through the lens of potential efficiencies, especially when they reduce double marginalization or improve coordination along the supply chain. In collusive settings, however, similar vertical tools can strengthen cartel conduct. Upstream participation can stabilize downstream collusion, weaken fringe competition, discipline retailers through wholesale terms, and transfer surplus between tiers in ways that are difficult to detect from retail prices alone. Treating cartels as a homogeneous category, or treating vertical participation as presumptively benign, may therefore understate harm when upstream firms actively facilitate downstream coordination.

Overall, the paper shows that the internal organization of a cartel matters for its economic effects. Vertical participation changes who joins the agreement, how deviations are disciplined, how prices are set along the supply chain, and how surplus is divided. These mechanisms help explain why vertical cartels generate larger retail-price and retail-markup effects than purely horizontal retail cartels in the Brazilian gasoline market. Recognizing vertical cartels as a distinct form of collusion is therefore important for measuring damages, designing remedies, and deterring coordinated conduct in vertically related markets.

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